

An Introduction to GNSS

GPS, GLONASS, Galileo and other Global Navigation Satellite Systems

NovAtel Inc.

First Edition



An Introduction to GNSS

GPS, GLONASS, Galileo and other
Global Navigation Satellite Systems

First Edition

Written by Charles Jeffrey for NovAtel Inc.

An Introduction to GNSS

GPS, GLONASS, Galileo and other Global Navigation Satellite Systems

First Edition

Written by Charles Jeffrey for NovAtel Inc.

Copyright 2010 NovAtel Inc.

Published by NovAtel Inc.

1120 - 68th Avenue N.E.

Calgary, Alberta, Canada

T2E 8S5

www.novatel.com

All rights reserved. Printed in Canada. No part of this book may be reproduced, stored in a retrieval system or transmitted in any form without prior written permission of the author and publisher except in the case of brief quotations in critical articles and reviews.

ISBN: 978-0-9813754-0-3

We acknowledge the contribution of Nicole Foss (project manager), Lori Winkler (editor), NovAtel Customer Support, and the many other employees at NovAtel who provided ideas and content for presenting complex GNSS concepts in simple, easy-to-understand terms.

We also acknowledge Mark Dulmadge of Storm Design Inc. for book design.

Table of Contents	iii
Foreword	2
Chapter 1 GNSS Overview	6
GNSS Systems	7
GPS (United States)	7
GLONASS (Russia)	7
Galileo (European Union)	7
Compass (China)	7
GNSS Architecture	7
Space Segment	8
Control Segment	9
User Segment	9
GNSS Signals	9
GNSS Positioning	9
GNSS Applications	11
GNSS User Equipment	11
GNSS Antennas	11
GNSS Receivers	12
GNSS Augmentation	12
Closing Remarks	12
Chapter 2 Basic GNSS Concepts	14
Step 1 - Satellites	15
Satellite Orbits	16
Satellite Signals	16
Satellite Errors	20
Satellite Lifetimes	20
Satellite Corrections	20
Step 2 - Propagation	20
Step 3 - Reception	21
Step 4 - Computation	22
GNSS Error Sources	25
Step 5 - Application	25
Advanced Concepts	26
Closing Remarks	26
Chapter 3 GNSS Satellite Systems	28
GPS (Global Positioning System, United States)	29
Space Segment	29
Signals	30

Control Segment	31
GPS Modernization	31
L2C	31
L5 (Safety of Life)	32
Other	32
GLONASS (Global Navigation Satellite System, Russia)	32
GLONASS System Design	33
GLONASS Space Segment	33
GLONASS Control Segment	34
GLONASS Signals	34
GLONASS and GPS	35
Galileo (European Union)	35
System Design	36
Compass (China)	37
IRNSS (Indian Regional Navigation Satellite System, India)	38
QZSS (Quasi-Zenith Satellite System, Japan)	38
Closing Remarks	38
Chapter 4 Advanced GNSS Concepts	40
Differential GNSS	40
Satellite-based Augmentation Systems	41
Free SBAS Services	42
Wide Area Augmentation System (WAAS)	42
European Geostationary Navigation Overlay Service (EGNOS)	43
CDGPS	43
Other SBAS Systems	44
Commercial SBAS Services	44
OmniSTAR	44
Real-Time Kinematic (RTK)	44
Virtual Reference Stations (VRS)	45
Dilution of Precision (DOP)	45
Combined GNSS/Inertial Navigation Systems	47
GNSS Data Post-Processing	48
Closing Remarks	49
Chapter 5 GNSS Applications and Equipment	52
Applications	52
Consumer	52
Transportation	53
Machine Control	53

Agriculture	54
Construction	55
Surface Mining	55
Surveying	55
Aerial Photogrammetry	56
Ground Mapping	56
Geospatial Information Systems (GIS)	57
Port Automation	57
Time Applications	58
Marine Applications	58
GNSS-Equipped Sonobuoys	58
Defence	59
Equipment	59
Closing Remarks	62
Afterword	64
Appendices	66
Appendix A – Acronyms	67
Appendix B – GNSS Glossary	71
Appendix C – Standards and References	83
Index	85

List of Figures:

Figure 1 Classroom Demonstration of GNSS Positioning	6
Figure 2 GNSS Segments	8
Figure 3 GNSS Satellite Orbits	8
Figure 4 Illustration of Trilateration – Knowing One Distance	9
Figure 5 Illustration of Trilateration – Knowing Two Distances	10
Figure 6 Illustration of Trilateration – Knowing Three Distances	10
Figure 7 GNSS User Equipment	11
Figure 8 GNSS Antennas	11
Figure 9 GNSS Receivers	12
Figure 10 Basic GNSS	14
Figure 11 Block IIR GPS Satellite	15
Figure 12 GPS Frequency Plan	17
Figure 13 Hedy Lamarr	18
Figure 14 GPS Navigation Message	19
Figure 15 GNSS Signal Propagation	20
Figure 16 GNSS Reception	21
Figure 17 Determining Time of Propagation	22
Figure 18 Ranging to First Satellite	23
Figure 19 Ranging to Second Satellite	23
Figure 20 Position Error	24
Figure 21 Detecting Position Error	24
Figure 22 Convergence	24
Figure 23 Launch of Galileo Satellite	28
Figure 24 GPS IIRM Satellite (artist’s rendition)	29
Figure 25 GPS Satellite Orbit	30
Figure 26 GPS Control Segment	31
Figure 27 GLONASS-M Satellite in Final Manufacturing	32
Figure 28 View of Earth (as seen by Apollo 17 crew)	33
Figure 29 GLONASS Antipodal Satellites	34
Figure 30 Galilei Galileo	35
Figure 31 Galileo Satellite in Orbit	36
Figure 32 Differential GNSS	40
Figure 33 SBAS – GNSS Data Gathering	41
Figure 34 SBAS – Correction Calculation and Uplink	42
Figure 35 SBAS – Correction Broadcast	42
Figure 36 Wide-Area Augmentation System (WAAS)	43
Figure 37 CDGPS Coverage	43

Figure 38 Real-Time Kinematic	45
Figure 39 Dilution of Precision (poor satellite geometry).	46
Figure 40 Dilution of Precision (improved geometry)	46
Figure 41 Minimizing Dilution of Precision	47
Figure 42 Combined GNSS/INS Solution.	48
Figure 43 Post Processing of GNSS Data.	49
Figure 44 Portable Navigation Device	52
Figure 45 Automatically Guided Trenching Machine	53
Figure 46 Precisely Aligned Crop Rows Planted by GNSS-guided Machinery	54
Figure 47 GNSS-based Surveying Equipment	55
Figure 48 Aerial Image of Niagara Falls.	56
Figure 49 GIS Data Output	57
Figure 50 Gantry Crane in the Port of Vancouver	57
Figure 51 Marine Application of GNSS.	58
Figure 52 Predator Unmanned Aerial Vehicle (UAV)	59
Figure 53 Examples of GNSS Equipment	60

List of Tables:

Table 1 GNSS System Errors	25
Table 2 GPS Satellite Constellation	29
Table 3 GPS Signal Characteristics	30
Table 4 GLONASS Satellite Constellation.	33
Table 5 GLONASS Signal Characteristics	34
Table 6 Galileo Satellite Constellation	36
Table 7 Galileo Services	37
Table 8 Planned Compass Satellite Constellation	37



Foreword

*They all laughed at Christopher Columbus
When he said the world was round.¹*

Christopher Columbus wasn't the first person to propose the world was round. Far from it. By the fifth century BCE, many Greek scholars had accepted a spherical earth as fact. Around 240 BCE, Eratosthenes, a Greek mathematician, poet, athlete, geographer and astronomer – a Renaissance man – ingeniously calculated the radius of the earth with surprisingly good accuracy.

Although Columbus knew the earth was round, he had obviously not read or agreed with Eratosthenes, for he significantly underestimated its size. He projected that heading west, the distance from the Canary Islands to Japan was 3,700 km, not 19,600 km as we now know it to be. Had Columbus known the true distance, he may have lost heart. He would certainly have had trouble convincing others to sail with him or to fund his first voyage.

Columbus navigated to the New World using dead reckoning, the technique of estimating one's current position based on a previously determined one – if I head west from a known location at 10 km/hr then, in two hours, I will be 20 km west of my starting point. The challenge in dead reckoning was the accurate and regular estimation of speed and heading.

A great deal of exploration was carried out at a time when positioning was not a very exact science, sometimes with dire consequences. In 1707, several ships of the Royal Navy struck the rocks near the Isles of Scilly, southwest of Cornwall, with a loss of four ships and 1400 men. Navigational error was blamed. Although it is not certain whether the error was in the determination of longitude or latitude, the tragedy led to the Longitude Act of 1714. Through this act, the British government offered prizes for people who could solve or advance the problem of accurately determining longitude, including one of £10,000 for a method that could determine longitude to within 60 nautical miles, about 111 kilometres.

Although none of the larger prizes offered by the Longitude Act were ever awarded, the initiative led to the development of many navigation techniques and equipment, including significant improvements in ship-borne chronometers (clocks), then critical to the accurate determination of longitude. With the advent of radio in the early 1900's, time signals were sent to ships, which could use the signal to regularly adjust their chronometers. In the 1940's, LORAN (LOng-RANge navigation system) was introduced. This allowed ships to triangulate their position using radio signals from LORAN stations at known shore-based locations. The first satellite, Sputnik, was launched in 1957 and it was not long before scientists contemplated working back from a known satellite orbit to determine a position on earth. Many of the problems faced by earlier navigators were quickly becoming historical footnotes.

¹ From "They All Laughed" by George and Ira Gershwin, a song popularized by, among others, Frank Sinatra and Ella Fitzgerald, in the 1950's.

What has most significantly changed navigation techniques is the advent of GNSS (Global Navigation Satellite Systems) which started with the launch of the U.S. Department of Defense Global Positioning System (GPS) in the late 1970's. Early applications of GNSS were developed for the military, and in the fields of surveying and mapping. Now, commercial operators of planes, trains and automobiles can know their position and heading quickly and accurately. Hikers can use handheld GNSS devices to calculate and display their position to within a few metres. Farmers can automatically guide their farm machinery for more efficient ploughing, seeding, fertilizing and harvesting. The world has changed.

The goal of this book is to present complex GNSS concepts and applications in a manner that informs without overwhelming. By the end of the book, you will understand the basics of GNSS and will have a foundation for further study or application.

Find a comfortable chair and a good reading light, and proceed to Chapter 1. We hope you enjoy the read.



Chapter 1

GNSS Overview

“New ideas pass through three periods: 1) It can’t be done. 2) It probably can be done, but it’s not worth doing. 3) I knew it was a good idea all along!” Arthur C. Clarke, British author, inventor and futurist.

Most of us now know that GNSS “was a good idea all along” and that we are now well into the third period. Although you might dare to argue that GNSS merits more than a hockey analogy, we will argue that understanding the basic concepts of satellite positioning is not much more difficult than understanding the rules of hockey. They are so straightforward, in fact, that one of our employees was asked by his daughter to explain it to her grade 4 class.

Before the class started, he set up the following demonstration, his version of “string theory.” He tacked cardboard figures of three satellites to the walls and ceiling of the classroom, as shown in Figure 1. Each “satellite” had a length of string stapled to it. He marked a location on the floor with a movable dot, then drew the strings down and marked where they all reached the dot. The strings now represented the distances from the dot to the individual satellites. He recorded the location of the dot and removed it from the floor.

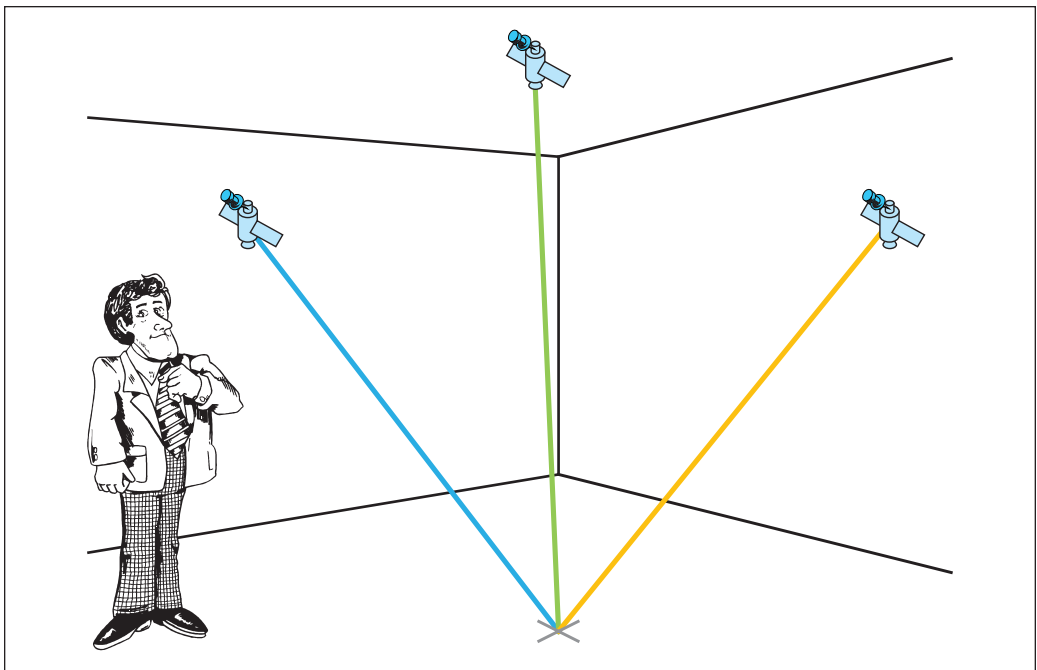


Figure 1 Classroom Demonstration of GNSS Positioning

When the students came into the classroom, our employee had them use the strings to determine the location. To do this, the students drew the strings down until the ends of the strings came together at one point on the floor. They marked this point with a movable dot and compared it with the previously marked position. They were very close. This simple demonstration showed that, if you know the location of three satellites and your distance from them, you can determine your position.

The determination of position is made quite a bit more complicated by several factors – the satellites are moving, the signals from the satellites are very weak by the time they reach the earth, the atmosphere interferes with the transmission of radio signals and, for cost reasons, the user equipment is not as sophisticated as the equipment in the satellites.

“The more you explain it, the more I don’t understand it.” Mark Twain, American author and humorist.

We agree. We will provide a more detailed explanation of position determination in Chapter 2.

GNSS Systems

Although you may already be familiar with the term “GPS” (Global Positioning System), you may not have heard the term “GNSS” (Global Navigation Satellite System), which is used to describe the collection of satellite positioning systems that are now operating or planned.

GPS (United States)

GPS was the first GNSS system and, as of today, it is the only one that is fully operational. GPS was launched in the late 1970’s by the United States Department of Defense. It now uses a constellation of between 24 and 32 satellites, and provides global coverage.

GLONASS (Russia)

GLONASS is operated by the Russian government. The full GLONASS constellation will consist of 24 satellites. Russia plans to provide global coverage by the end of 2010.

Galileo (European Union)

Thirty satellites are planned with the first being launched in 2006. The full constellation will not be complete for several years.

Compass (China)

Compass will be the Chinese navigation satellite system. The system will consist of 35 satellites. A regional service will be provided by 2010 then the service will be extended to provide global coverage in the years 2015-2020.

In Chapter 3, we will provide additional information about these systems. As GNSS constellations and satellites are added, we will be able to calculate position more accurately and in more and more places.

GNSS Architecture

“The future ain’t what it used to be.” Yogi Berra, former Major League Baseball player and manager.

Mr. Berra is correct. The implementation of GNSS satellite systems has really changed things.

GNSS satellite systems consist of three major components or “segments”: space segment, control segment and user segment. These are illustrated in Figure 2.

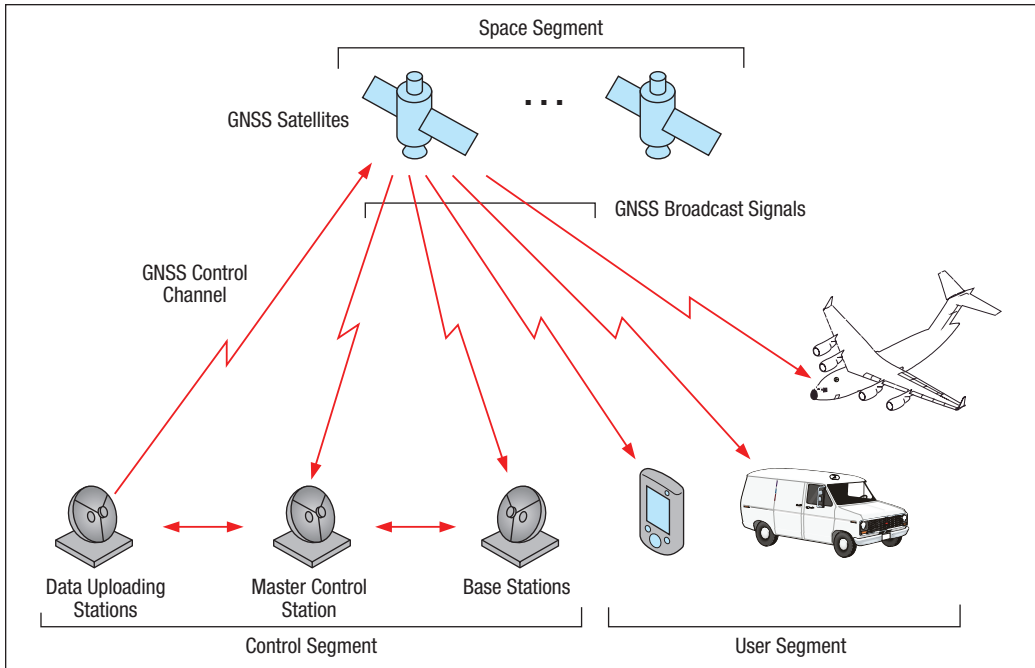


Figure 2 GNSS Segments

Space Segment

The space segment consists of GNSS satellites, orbiting about 20,000 km above the earth. Each GNSS has its own “constellation” of satellites, arranged in orbits to provide the desired coverage, as illustrated in Figure 3.

Each satellite in a GNSS constellation broadcasts a signal that identifies it and provides its time, orbit, and status. To illustrate, consider the following. You are downtown. You call a friend. Your friend is not at home, so you leave a message:

This is Charles [identity]. The time is 1:35 PM [time]. I am at the northwest corner of 1st Avenue and 2nd Street and I am heading towards your place [orbit]. I am OK, but I am a bit thirsty [status].

Your friend returns a couple of minutes later, listens to your message and “processes” it, then calls you back and suggests you come up a slightly different way; effectively, your friend has given you an “orbit correction.”

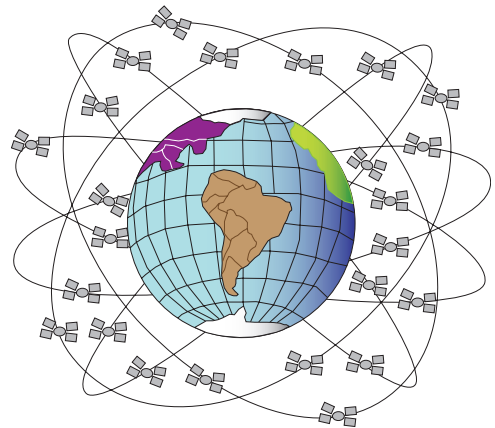


Figure 3 GNSS Satellite Orbits

Control Segment

The control segment comprises a ground-based network of master control stations, data uploading stations, and monitor stations; in the case of GPS, two master control stations (one primary and one backup), four data uploading stations and ten monitor stations, located throughout the world.

In each GNSS system, the master control station adjusts the satellites' orbit parameters and onboard high-precision clocks when necessary to maintain accuracy.

Monitor stations, usually installed over a broad geographic area, monitor the satellites' signals and status, and relay this information to the master control station. The master control station analyses the signals then transmits orbit and time corrections to the satellites through data uploading stations.

User Segment

The user segment consists of equipment that processes the received signals from the GNSS satellites and uses them to derive and apply location and time information. The equipment ranges from handheld receivers used by hikers, to sophisticated, specialized receivers used for high-end survey and mapping applications.

GNSS Signals

GNSS radio signals are quite complex. Their frequencies are around 1.5 GHz (gigahertz) – 1.5 billion cycles per second. GNSS operates at frequencies that are higher than FM radio, but lower than a microwave oven. By the time GNSS signals reach the ground, they are very, very weak. We will provide more information about how the user segment deals with this in Chapter 2.

GNSS Positioning

"I have never been lost, but I will admit to being confused for several weeks." Daniel Boone, American pioneer and hunter.

If you have a GNSS receiver, it is unlikely that you will ever be lost again. GNSS positioning is based on a process called "trilateration." Simply put, if you don't know your position, but do know your distance from three known points, you can trilaterate your location.

Let's say you are 3 km from Person A's house. All you know is that you are on a circle 3 km from Person A's, as shown in Figure 4.

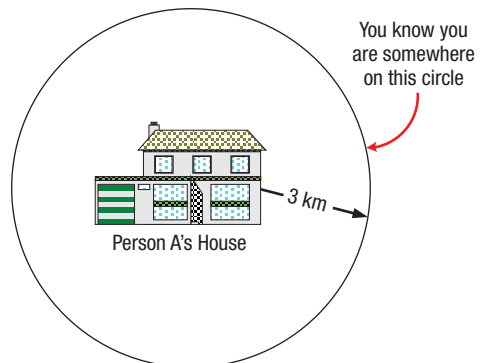


Figure 4 Illustration of Trilateration – Knowing One Distance

But if you also know that you are 4 km from Person B's house, you will have a much better idea of where you are, since only two places (x and y) exist on both circles, as shown in Figure 5.

With a third distance, you can only be in one physical location. If you are 6 km from Person C's house, you have to be at position x since this is the only location where all three circles (distances) meet.

In Chapter 2, we will show you how the technique of trilateration is extended to GNSS. Conceptually, we are just going to extend the above example by replacing the houses with satellites. And for reasons that we will outline, we will replace the three houses with four satellites.

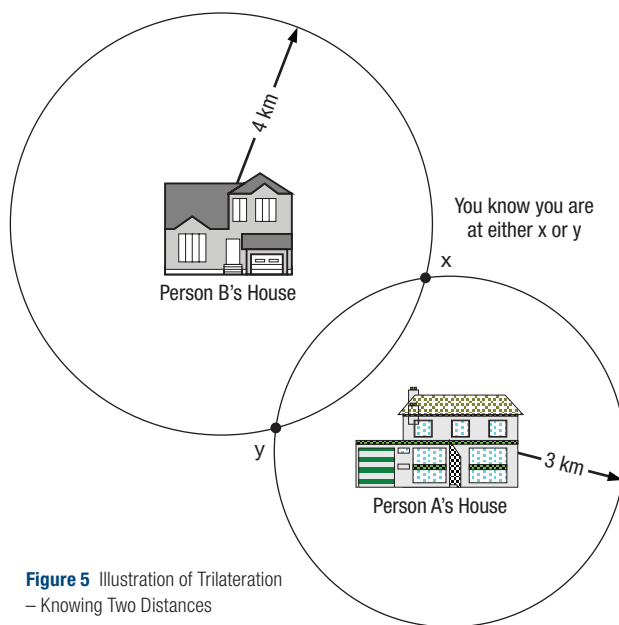


Figure 5 Illustration of Trilateration – Knowing Two Distances

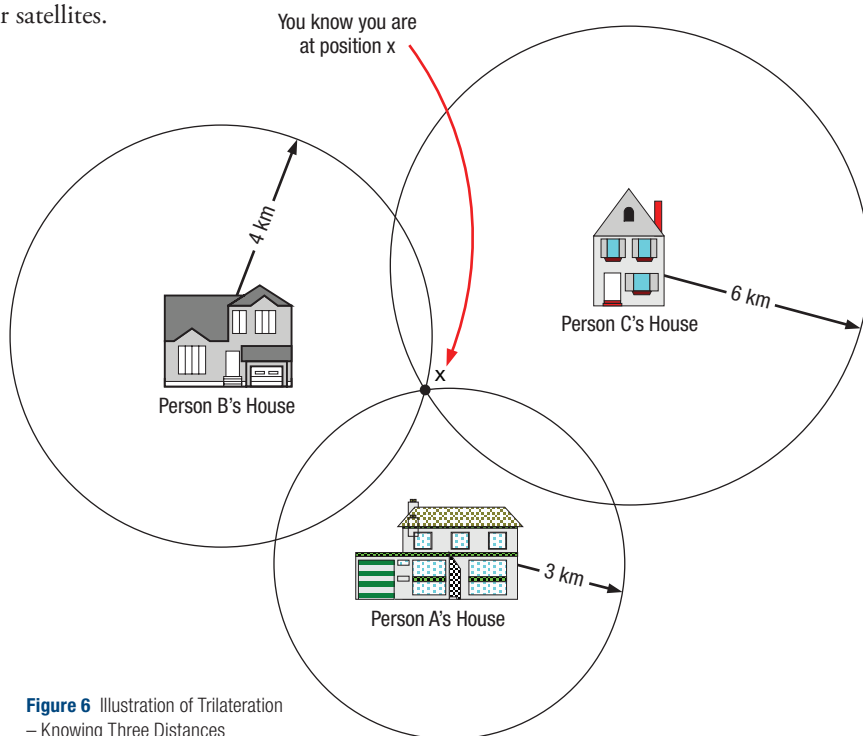


Figure 6 Illustration of Trilateration – Knowing Three Distances

GNSS Applications

The first non-military applications of GNSS technology were in surveying and mapping. Today, GNSS is being used for commercial applications in agriculture, transportation, machine control, marine navigation, and other industries where efficiencies can be gained from the application of precise, continually available position and time information. GNSS is also used in a broad range of consumer applications, including vehicle navigation, mobile communications, hiking and athletics. As GNSS technology improves and becomes less expensive, more and more applications will be conceived and developed.

In addition to position, GNSS receivers can provide users with very accurate time, by “synchronizing” their local clock with the high-precision clocks onboard the satellites. This has enabled technologies and applications such as the synchronization of power grids, cellular systems, the internet and financial networks.

We’ll talk more about GNSS applications in Chapter 5.

GNSS User Equipment

The primary components of the GNSS user segment are antennas and receivers, as shown in Figure 7. Depending on the application, antennas and receivers may be physically separate or they may be integrated into one assembly.

GNSS Antennas

GNSS antennas receive the radio signals that are transmitted by the GNSS satellites, and send these signals to the receivers. GNSS antennas are available in a range of shapes, sizes, and performances. The antenna is selected based on the application. While a large antenna may be appropriate for a base station, a low-profile aerodynamic antenna may be more suitable for aircraft installations. Figure 8 presents a sampling of GNSS antennas.

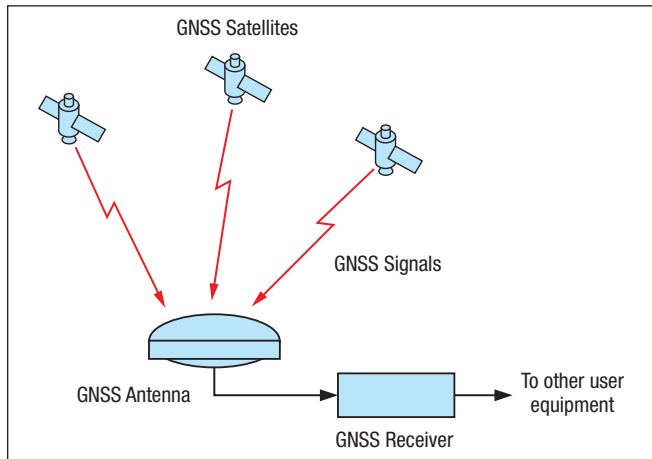


Figure 7 GNSS User Equipment



Figure 8 GNSS Antennas

GNSS Receivers

Receivers process the satellite signals recovered by the antenna to calculate position and time. Receivers may be designed to use signals from one GNSS constellation or from more than one GNSS constellation. As with antennas, and as illustrated in Figure 9, receivers may be packaged for a particular application, such as aviation or agriculture.

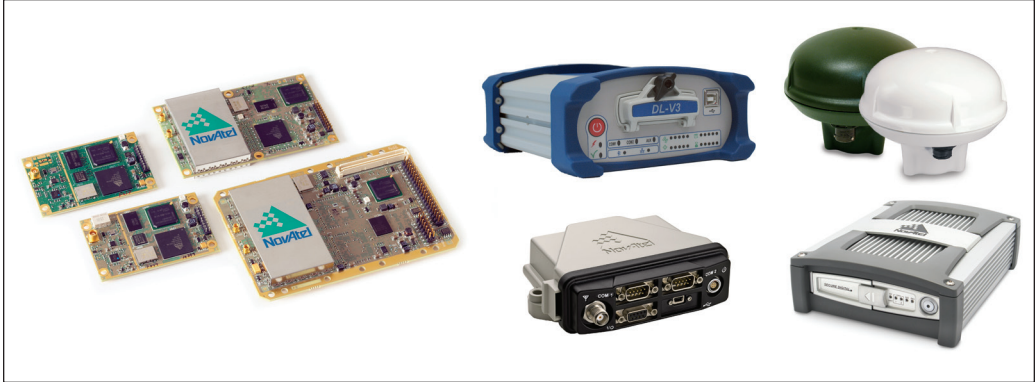


Figure 9 GNSS Receivers

We will talk more about GNSS equipment in Chapter 5.

GNSS Augmentation

Positioning based on standalone GNSS service is accurate to within a few metres. The accuracy of standalone GNSS, and the number of available satellites, may not be adequate for the needs of some users.

Techniques and equipment have been developed to improve the accuracy and availability of GNSS position and time information. We will discuss some of these techniques in Chapter 4.

Closing Remarks

Chapter 1 provided an overview of the main concepts and components of GNSS. This high-level summary will help your understanding as we present GNSS in greater detail, starting with a more thorough look at basic GNSS concepts in Chapter 2.



Chapter 2

Basic GNSS Concepts

In this chapter, we will introduce basic GNSS concepts. We'll discuss more advanced concepts in Chapter 4.

"Any sufficiently advanced technology is indistinguishable from magic." Arthur C. Clarke.

GNSS may at first seem like magic, but the more you study and learn about it, the simpler and more elegant it will become. The basic GNSS concept is shown in Figure 10, which illustrates the steps involved in using GNSS to determine time and position then applying this information.

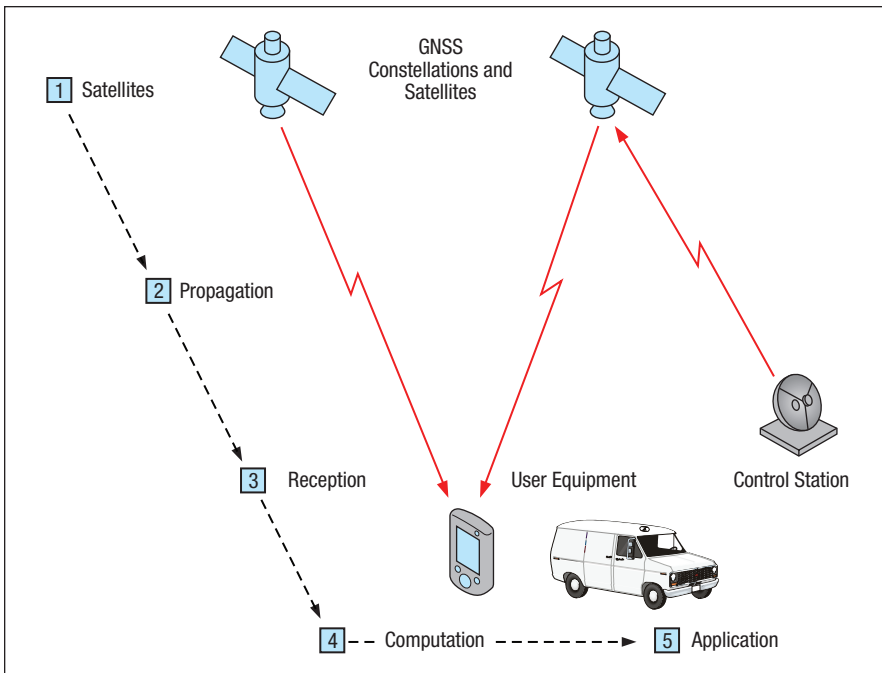


Figure 10 Basic GNSS

Step 1 - Satellites: GNSS satellites orbit the earth. The satellites know their orbit ephemerides (the parameters that define their orbit) and the time very, very accurately. Ground-based control stations adjust the satellites' ephemerides and time, when necessary.

Step 2 - Propagation: GNSS satellites regularly broadcast their ephemerides and time, as well as their status. GNSS radio signals pass through layers of the atmosphere to the user equipment.

Step 3 - Reception: GNSS user equipment receives the signals from multiple GNSS satellites then, for each satellite, recovers the information that was transmitted and determines the time of propagation, the time it takes the signals to travel from the satellite to the receiver.

Step 4 - Computation: GNSS user equipment uses the recovered information to compute time and position.

Step 5 - Application: GNSS user equipment utilizes the position and time information in their applications, for example, navigation, surveying or mapping.

In the following sections, we will discuss each of the above steps in more detail.

Step 1 - Satellites

There are multiple constellations of GNSS satellites orbiting the earth. A constellation is simply an orderly grouping of satellites, typically 20-30, in orbits that have been designed to provide a desired coverage, for example, regional or global. We will provide more details about GNSS constellations in Chapter 3.

GNSS satellites orbit well above the atmosphere, about 20,000 km above the earth's surface. They are moving very fast, several kilometres per second.

GNSS satellites are not as small as you might think. The latest generation of GPS satellites weighs over 1000 kg, a bit more than the weight of a Volkswagen Beetle. The first generation of Galileo satellites will weigh about 700 kg, and the body of these satellites will be 2.7 m x 1.2 m x 1.1 m. When the satellite's solar arrays (the large panels that collect the sun's energy and use it to power the satellite) are deployed, they will be over 18 m in length. Figure 11 shows a picture of the body of a Block IIR GPS satellite, to give a sense of how large they are.



Lockheed Martin

Figure 11 Block IIR GPS Satellite

In the relative vacuum of space, satellite trajectories are very stable and predictable. As mentioned, GNSS satellites know their time and orbit ephemerides very, very accurately. If you ask a GPS satellite for the time, it won't tell you eight thirty. It will tell you 8:31.39875921.

The latest generation of GPS satellites uses rubidium clocks that are accurate to within ± 5 parts in 10^{11} . These clocks are synchronized by more-accurate ground-based cesium clocks. You would need to watch one of these cesium clocks for over 100,000 years to see it gain or lose a second. By comparison, if you have a quartz watch, it will likely have an accuracy of ± 5 parts in 10^6 and will lose about a second every two days.

By the way, if all GNSS receivers needed a rubidium standard, the viability of GNSS would quickly collapse. Later in the chapter, we will describe the elegant way GNSS systems “transfer” the accuracy of the satellite clocks to GNSS receivers.

You may be wondering why time is such a big deal in GNSS systems. It is because the time it takes a GNSS signal to travel from satellites to receivers is used to determine distances (ranges) to satellites. Accuracy is required because radio waves travel at the speed of light. In one microsecond (a millionth of a second), light travels 300 m. In a nanosecond (a billionth of a second), light travels 30 cm. Small errors in time can result in large errors in position.

GPS was the first GNSS constellation to be launched. At a cost of US\$12 billion, it is the most accurate navigation system in the world. The Russian GLONASS constellation has also been launched and is operational. The benefit to end users of having access to multiple constellations is redundancy and availability. If one system fails, for any reason, GNSS receivers, if they are equipped to do so, can receive and use signals from satellites in other systems. System failure does not happen often, but it is nice to know that if it did, your receiver may still be able to operate.

Regardless, access to multiple constellations is of particular benefit where line of sight to some of the satellites is obstructed, as is often the case in urban or foliated areas.

Satellite Orbits

GNSS satellites orbit well above the Earth's atmosphere. GPS and GLONASS satellites orbit at altitudes close to 20,000 km. Galileo satellites will orbit a bit higher, around 23,000 km. GNSS orbits, which are more or less circular, and highly stable and predictable, fall into the category of MEO, for medium earth orbit.

There is not much drag at 20,000 km, but gravitational effects and the pressure of solar radiation do affect GNSS orbits a bit and the orbits have to be occasionally corrected. While its orbit is being adjusted, a GNSS satellite's status is changed to “out of service” so user equipment knows not to use their signals.

Satellite Signals

“Everything should be made as simple as possible, but no simpler.” Albert Einstein.

GNSS satellite signals are complex. Describing these signals requires equally complex words like pseudorandom, correlation, and code division multiple access (CDMA). To explain these GNSS concepts, let's first discuss GPS satellite signals.

First and foremost, GPS was designed as a positioning system for the US Department of Defense. To provide high-accuracy position information for military applications, a lot of complexity was designed into the system to make it secure and impervious to jamming and interference. Although military and civilian components of GPS are separate, some of the technologies used in the military component have been applied to the civilian component.

Since it achieved initial operational capability in December 1993, GPS has been available to civilian users, who have different requirements for service availability, positioning accuracy and cost.

The frequency plans (plans that describe the frequency, amplitude and width of signals) for each GNSS system are a little different. We will describe these plans in more detail in Chapter 3. To illustrate GNSS concepts, however, we will briefly describe the frequency and signal scheme used by GPS, which is shown in Figure 12. Conceptually, this is not much different than the frequency plan for cable or broadcast television channels.

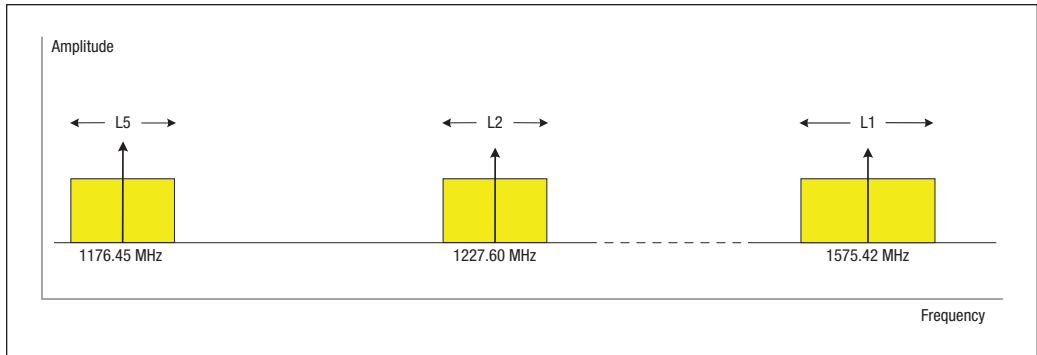


Figure 12 GPS Frequency Plan

As shown in Figure 12, GPS satellites transmit information on the L1, L2 and L5 frequencies. You may ask, “How can all GPS satellites transmit on the same frequencies?”

GPS works the way it does because of the transmission scheme it uses, which is called CDMA. CDMA is a form of spread spectrum. GPS satellite signals, although they are on the same frequency, are modulated by a unique pseudorandom digital sequence, or code. Each satellite uses a different pseudorandom code. Pseudorandom means that the signal only appears random; in fact, it actually repeats after a period of time. Receivers know the pseudorandom code for each satellite. This allows receivers to correlate (synchronize) with the CDMA signal for a particular satellite. CDMA signals are at a very low level, but through this code correlation, the receiver is able to recover the signals and the information they contain.

To illustrate, consider listening to a person in a noise-filled room. Many conversations are taking place, but each conversation is in a different language. You are able to understand the person because you know the language they are speaking. If you are multilingual, you will be able to understand what other people are saying too. CDMA is a lot like this.

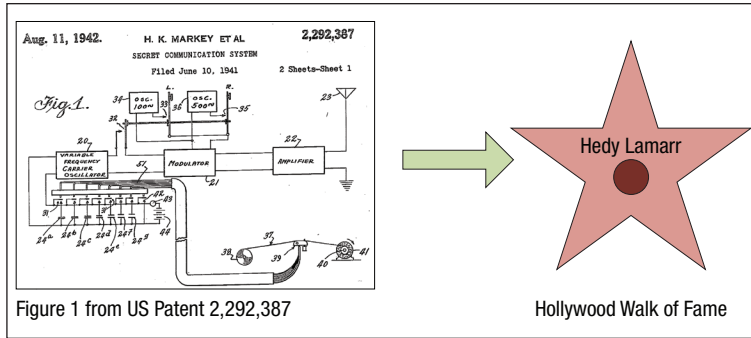


Figure 13 Hedy Lamarr

You might be interested to learn that Hedy Lamarr, Austrian-born American scientist and actress, co-invented an early form of spread-spectrum communications technology. On August 11, 1942, she and her co-worker, George Antheil, were granted U.S. Patent 2,292,387. Unbelievably, Lamarr shifted careers and went on to make 18 films from 1940 to 1949, but the concepts covered in her patent contributed to the development of today's spread spectrum communications.

GPS operates in a frequency band referred to as the L-band, a portion of the radio spectrum between 1 and 2 GHz. L-band was chosen for several reasons, including:

- Simplification of antenna design. If the frequency had been much higher, user antennas may have had to be a bit more complex.
- Ionospheric delay is more significant at lower frequencies. We'll talk more about ionospheric delay in Step 2 - Propagation.
- Except through a vacuum, the speed of light is lower at lower frequencies, as evident by the separation of the colours in light by a prism. You may have thought the speed of light was a constant at 299,792,458 metres per second. It is actually 299,792,458 metres per second in a vacuum, but through air or any other medium, it is less.
- The coding scheme requires a high bandwidth, which was not available in every frequency band.
- The frequency band was chosen to minimize the effect that weather has on GPS signal propagation.

L1 transmits a navigation message, the coarse acquisition C/A code (freely available to the public) and an encrypted precision (P) code, called the P(Y) code (restricted access). The navigation message is a low bit rate message that includes the following information:

- GPS date and time.
- Satellite status and health. If the satellite is having problems or its orbit is being adjusted, it will not be usable. When this happens, the satellite will transmit the out-of-service message.
- Satellite ephemeris data, which allows the receiver to calculate the satellite's position. This information is accurate to many, many decimal places. Receivers can determine exactly where the satellite was when it transmitted its time.

- Almanac, which contains information and status for all GPS satellites, so receivers know which satellites are available for tracking. On start up, a receiver will recover this “almanac.” The almanac consists of coarse orbit and status information for each satellite in the constellation.

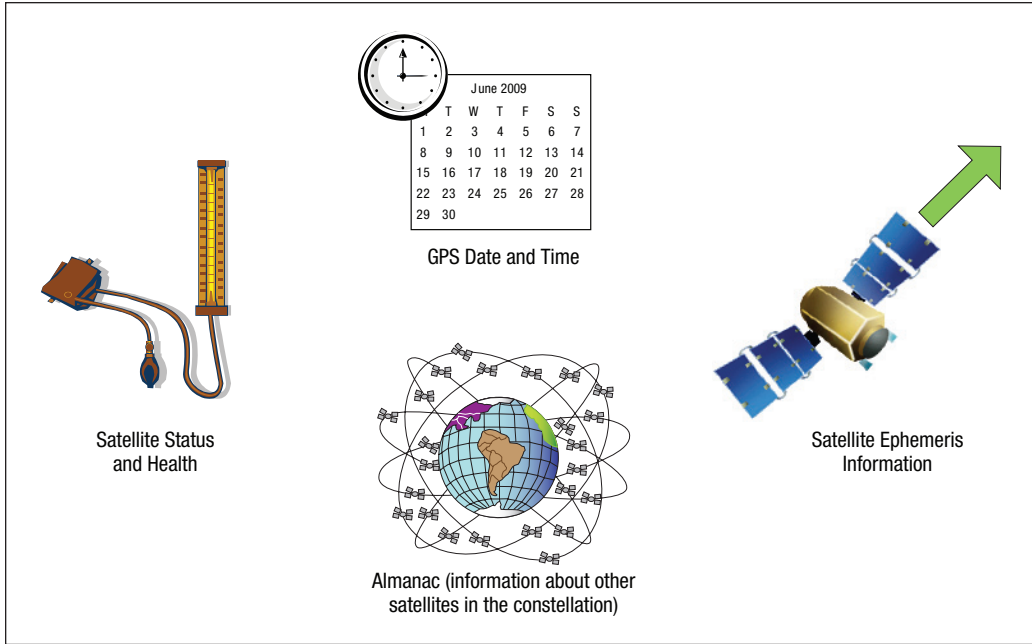


Figure 14 GPS Navigation Message

The P(Y) code is for military use. It provides better interference rejection than the C/A code, which makes military GPS more robust than civilian GPS. The L2 frequency transmits the P(Y) code and, on newer GPS satellites, it also transmits the C/A code (referred to as L2C), providing a second publicly available code to civilian users. Although the information in the P(Y) code is not accessible to everyone, clever people have figured out ways to use the L2 carrier and code, without knowing how it is coded.

While the GPS transmission scheme is complex, it was chosen for many good reasons:

- GPS receivers can recover very weak signals using very small antennas. This keeps the receiver cost low.
- Multi-frequency operation allows for ionospheric compensation, since ionospheric delays vary with frequency.
- The GPS system is resistant to jamming and interference.
- Security. Signals accessed and used by military applications are not accessible by civilians.

Other GNSS systems are conceptually similar to GPS, but there are differences. We will provide more information about these differences in Chapter 3.

Satellite Errors

Satellite errors include ephemeride and clock errors. These satellite errors are very, very small, but keep in mind that in one nanosecond, light travels 30 centimetres.

Satellite Lifetimes

GNSS satellites don't last forever. Sometimes they are phased out with newer models that have new signals or improved time keeping. Sometimes GNSS satellites do fail and, if they can't be restored, are permanently removed from service.

Satellite Corrections

Earth stations continuously monitor the satellites and regularly adjust their time and orbit information, to keep this broadcasted information highly accurate. If a satellite's orbit drifts outside the operating limits, it may be taken out of service, and its orbit adjusted using small rocket boosters.

In our step-by-step illustration of GNSS, the radio signals have left the satellite antenna and are hurtling earthbound at the speed of light.

Step 2 - Propagation

GNSS signals pass through the near-vacuum of space, then through the various layers of the atmosphere to the earth, as illustrated in Figure 15.

To obtain accurate positioning and timing, we need to know the length of the direct path from the satellite to the user equipment (which we refer to as the “range” to the satellite). As shown in Figure 15, radio waves do not travel in a straight path. Light travels in a straight line only in a vacuum or through a perfectly homogeneous medium. Just as a straw is seemingly “bent” in a glass of water, radio signals from the satellite are bent as they pass through the earth's atmosphere. This “bending” increases the amount of time the signal takes to travel from the satellite to the receiver. As we shall explain in Step 4, the distance to the satellite is calculated by multiplying the time of propagation (which, you recall, is the time it takes the signals to travel from the satellite to the receiver) by the speed of light. Errors in the propagation time increase or decrease the computed range to the satellite. Incidentally, since the computed range contains errors and is not exactly equal to the actual range, we refer to it as a “pseudorange.”

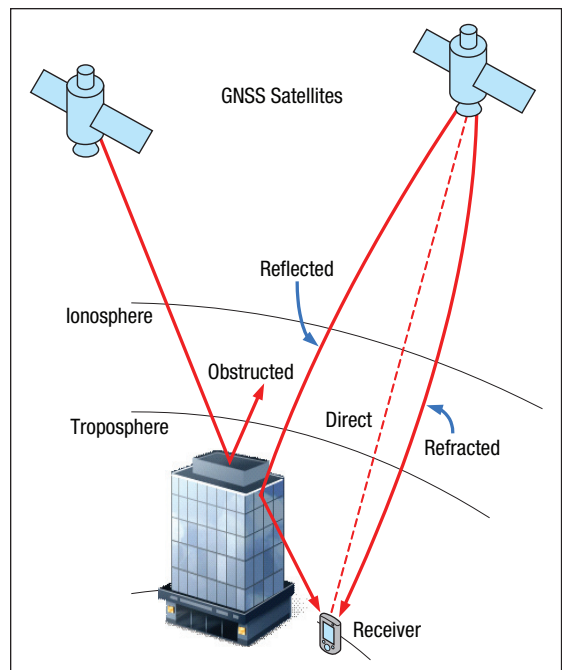


Figure 15 GNSS Signal Propagation

The layer of the atmosphere that most influences the transmission of GPS (and other GNSS) signals is the ionosphere, the layer 70 to 1000 km above the earth's surface. Ultraviolet rays from the sun ionize gas molecules in this layer, releasing free electrons. These electrons influence electromagnetic wave propagation, including GPS satellite signal broadcasts. Ionospheric delays are frequency dependent so by calculating the range using both L1 and L2, the effect of the ionosphere can be virtually eliminated by the receiver.

The other layer of the atmosphere that influences the transmission of GPS signals is the troposphere, the lowest layer of the Earth's atmosphere. The thickness of the troposphere varies, about 17 km in the middle latitudes, up to 20 km nearer the equator, and thinner at the poles. Tropospheric delay is a function of local temperature, pressure and relative humidity. L1 and L2 are equally delayed, so the effect of tropospheric delay cannot be eliminated the way ionospheric delay can be. It is possible, however, to model the troposphere then predict and compensate for much of the delay.

Some of the signal energy transmitted by the satellite is reflected on the way to the receiver. This phenomenon is referred to as “multipath propagation.” These reflected signals are delayed from the direct signal and, if they are strong enough, can interfere with the desired signal. Techniques have been developed whereby the receiver only considers the earliest-arriving signals and ignores multipath signals, which arrive later. In the early days of GPS, most errors came from ionospheric and tropospheric delays, but now more attention is being made to multipath effects, in the interests of continually improving GNSS performance.

Step 3 Reception

As we have indicated, receivers need at least four satellites to obtain a position. The use of more satellites, if they are available, will improve the position solution; however, the receiver's ability to make use of additional satellites may be limited by its computational power. The manner by which the receiver uses the additional ranges will generally be the intellectual property of the manufacturer.

Depending on the implementation, user equipment can recover signals from multiple satellites in multiple GNSS constellations.

To determine a fix (position) and time, GNSS receivers need to be able to track at least four satellites. This means there needs to be a line of sight between the receiver's antenna and the four satellites.

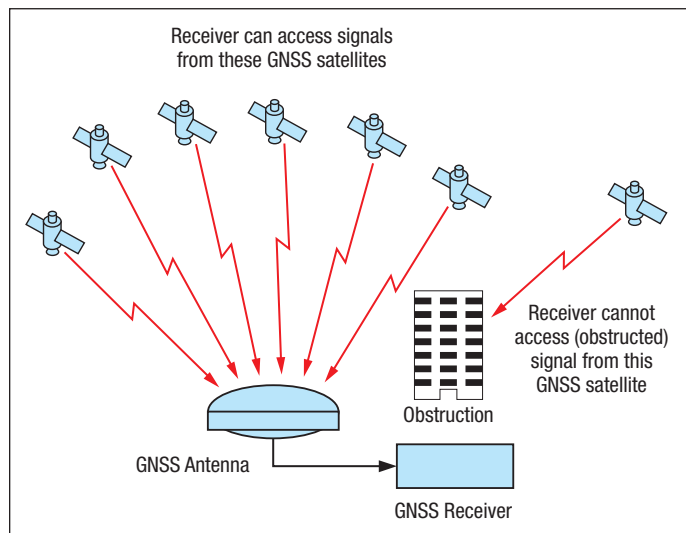


Figure 16 GNSS Reception

Receivers vary in terms of which constellation or constellations they track, and how many satellites they track simultaneously.

For each satellite being tracked, the receiver determines the propagation time. It can do this because of the pseudorandom nature of the signals. To illustrate, refer to Figure 17, which shows the transmission of a pseudorandom code, a series of zeroes and ones. Since the receiver knows the pseudorandom code for each satellite, it can determine the time it received the code from a particular satellite. In this way, it can determine the time of propagation, as shown.

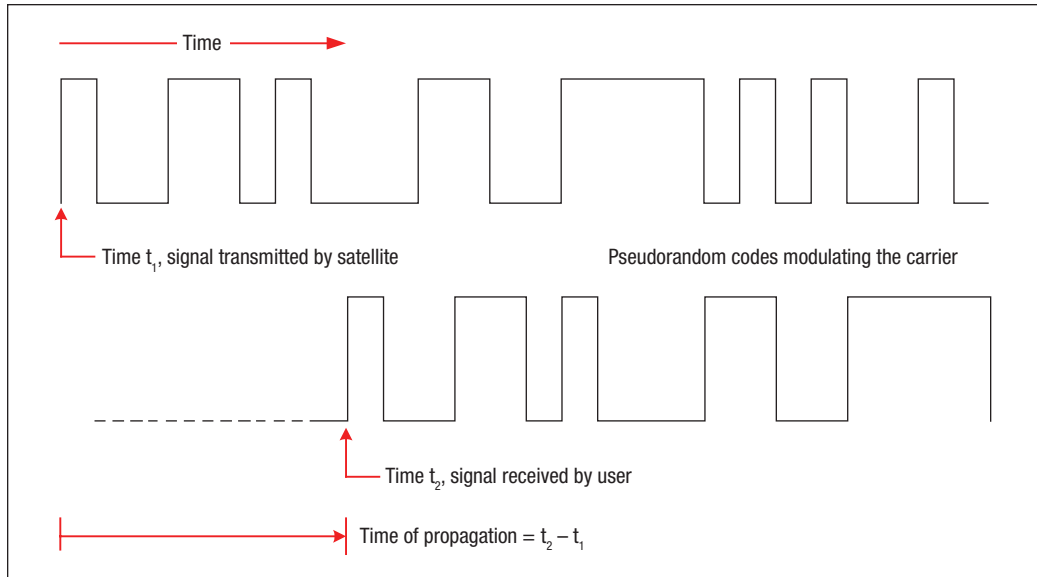


Figure 17 Determining Time of Propagation

Step 4 Computation

If we knew the exact position of three satellites and the exact range to each of them, we would geometrically be able to determine our location. We have suggested that we need ranges to four satellites to determine position. In this section, we will explain why this is so, and how GNSS positioning actually works.

For each satellite being tracked, the receiver calculates how long the satellite signal took to reach it, as follows:

$$\text{Propagation Time} = \text{Time Signal Reached Receiver} - \text{Time Signal Left Satellite}$$

Multiplying this propagation time by the speed of light gives the distance to the satellite.

For each satellite being tracked, the receiver now knows where the satellite was at the time of transmission (because the satellite broadcasts its orbit ephemerides) and it has determined the distance to the satellite when it was there. Using trilateration, a method of geometrically determining the position of an object, in a manner similar to triangulation, the receiver calculates its position.

To help us understand trilateration, we'll present the technique in two dimensions. The receiver calculates its range to Satellite A. As we mentioned, it does this by determining the amount of time it took for the signal from Satellite A to arrive at the receiver, and multiplying this time by the speed of light. Satellite A communicated its location (determined from the satellite orbit ephemerides and time) to the receiver, so the receiver knows it is somewhere on a circle with radius equal to the range and center at the location of Satellite A, as illustrated in Figure 18. In three dimensions, we would show ranges as spheres, not circles, but stay with us for now.

The receiver also determines its range to a second satellite, Satellite B. Now the receiver knows it is at the intersection of two circles, at either Position 1 or 2, as shown in Figure 19.

By knowing the location of Satellite A and your distance to it, you know you are somewhere on this circle

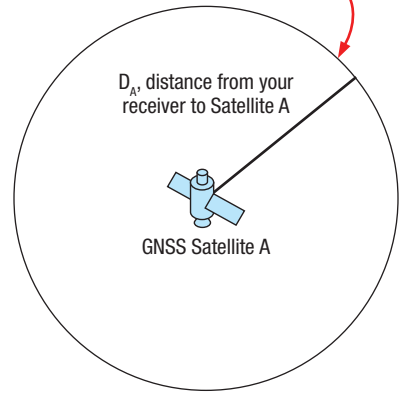


Figure 18 Ranging to First Satellite

By knowing the location of Satellites A and B, and your distance to each of them, you know you are at Position 1 or Position 2

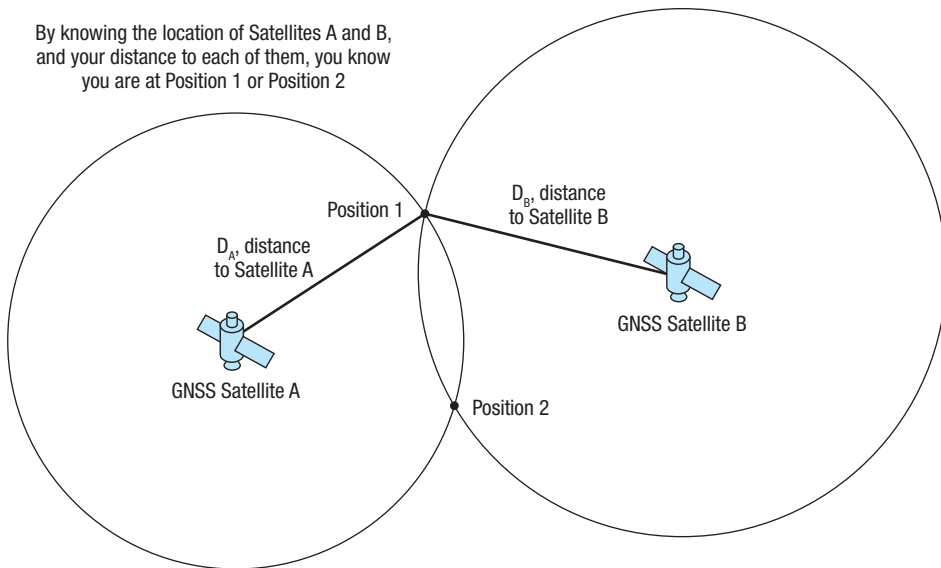


Figure 19 Ranging to Second Satellite

You may be tempted to conclude that ranging to a third satellite would be required to resolve your location to Position 1 or Position 2. But one of the positions can most often be eliminated as not feasible because, for example, it is in space or in the middle of the Earth. You might be tempted to extend our illustration to three dimensions and suggest that only three ranges are needed for positioning.

But as we discussed earlier, four ranges are necessary. Why is this?

It turns out that receiver clocks are not nearly as accurate as the clocks on board the satellites. Most are based on quartz crystals. Remember, we said these clocks were accurate to only about 5 parts per million. If we multiply this by the speed of light, it will result in an accuracy of ± 1500 metres. When we determine the range to two satellites, our computed position will be out by an amount proportional to the inaccuracy in our receiver clock, as illustrated in Figure 20.

We want to determine our actual position but, as shown in Figure 20, the receiver time inaccuracy causes range errors that result in position errors. The receiver knows there is an error, it just does not know the size of the error. If we now compute the range to a third satellite, it will not intersect the computed position, as shown in Figure 21.

Now for one of the ingenious techniques used in GNSS positioning.

The receiver knows that the reason the pseudoranges to the three satellites are not intersecting is because its clock is not very good. The receiver is programmed to advance or delay its clock until the pseudoranges to the three satellites converge at a single point, as shown in Figure 22.

The incredible accuracy of the satellite clock has now been “transferred” to the receiver clock, eliminating the receiver clock error in the position determination. The receiver now has both an accurate position and a very, very accurate time. This presents opportunities for a broad range of applications, as we shall discuss.

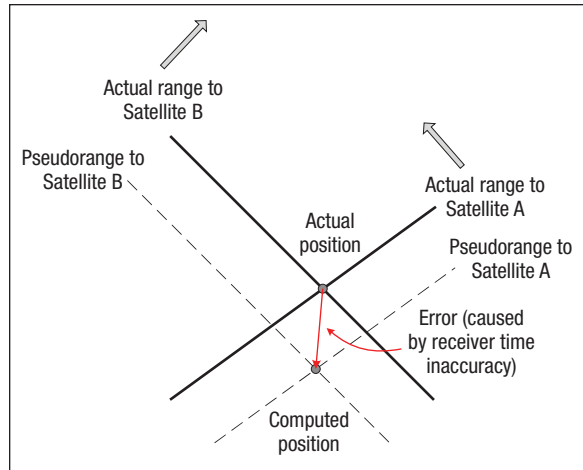


Figure 20 Position Error

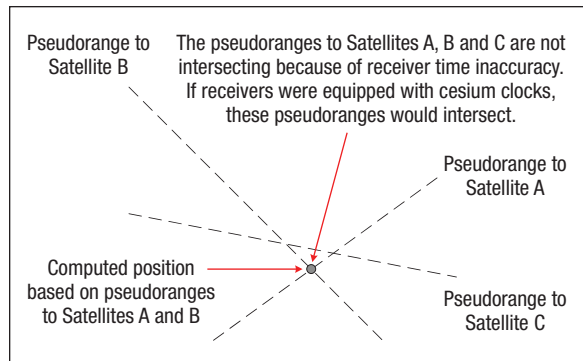


Figure 21 Detecting Position Error

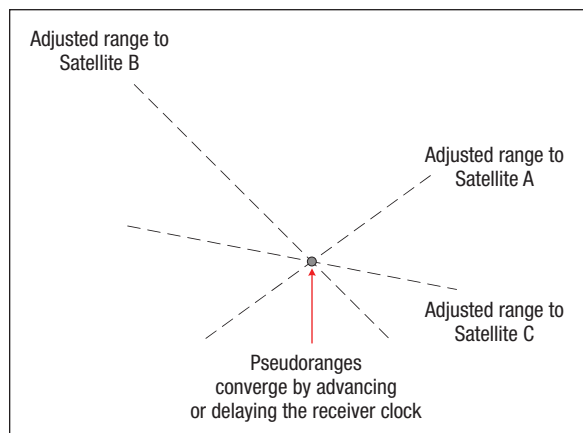


Figure 22 Convergence

The above technique shows how, in a two-dimensional representation, receiver time inaccuracy can be eliminated and position determined using ranges to three satellites. When we extend this technique to three dimensions, we need to add a range to a fourth satellite. This is the reason why line-of-sight to a minimum of four GNSS satellites is needed to determine position.

GNSS Error Sources

Before we move on, let's summarize the errors that can affect the accuracy of standard GNSS pseudorange determination, that is, the determination of the pseudorange to a single satellite. These are shown in Table 1:

Contributing Source	Error Range
Satellite clocks	± 2 m
Orbit errors	± 2.5 m
Ionospheric delays	± 5 m
Tropospheric delays	± 0.5 m
Receiver noise	± 0.3 m
Multipath	± 1 m

Table 1 GNSS System Errors

The degree with which the above pseudorange errors affect positioning accuracy depends largely on the geometry of the satellites being used. In Chapter 4, we'll discuss techniques for reducing these errors further.

Step 5 Application

Once the receiver has determined its position and time, this information is passed to and used by the user application. In a handheld receiver, for example, position may be superimposed on a map to graphically show the user's location. Events in electrical substations can be tagged with time that is exactly the same for two remote substations. Well, within nanoseconds anyway.

The GNSS market is growing rapidly and is already a multi-billion dollar industry. We will discuss GNSS applications in more detail in Chapter 5.

Advanced Concepts

In this chapter, we have described the basic concepts of GNSS positioning. As we noted in Chapter 1, very smart people have developed techniques for extending and improving the achievable accuracy. We will discuss some of these techniques in Chapter 4:

- **Dilution of Precision:** The geometric arrangement of satellites, as they are presented to the receiver, affects the accuracy of the position and time solution. Receivers will use the signals from satellites to minimize this “dilution of precision.”
- **Differential GNSS:** A technique whereby a base station at a precisely known location works backwards to determine errors in ranges to available GPS satellites, then transmits range corrections to other stations. This technique is used to eliminate or reduce satellite errors and atmospheric delays.
- **Carrier-based Techniques:** The positioning technique we have described in this chapter is referred to as a code-based technique because the receiver correlates with and uses the pseudorandom codes transmitted by four or more satellites to determine its position and time. This results in positioning accuracies of a few metres. For applications such as surveying, higher accuracies are required. Carrier-based techniques such as Real-Time Kinematic (RTK) have been developed that can provide positions that are orders of magnitude more accurate than the code-based GNSS we have discussed in this chapter.
- **Inertial Navigation Systems:** In mobile applications, line of sight to GNSS satellites may be interrupted when, for example, a vehicle is passing through a tunnel or built-up urban area. GNSS receivers may be integrated with an inertial navigation system to bridge through periods where GNSS service is not available.

Closing Remarks

This has been a tough chapter and we're pleased you persevered through the basics of GNSS positioning. Chapter 3 provides additional information about the GNSS constellations that have been implemented or are planned. Chapter 4 discusses advanced GNSS concepts, and Chapter 5 discusses equipment and applications – how the simple outputs of this incredible technology are being used.



Chapter 3

GNSS Satellite Systems

“The dinosaurs became extinct because they didn’t have a space program.” Larry Niven, American science fiction author.

Larry Niven is suggesting is that if the dinosaurs had had a space program, they could have intercepted and deflected the asteroid that some think may have hit the earth and led to the extinction of the dinosaurs.

Unlike the dinosaurs, several countries now have existing or planned space programs that include the implementation of national or regional Global Navigation Satellite Systems. In this chapter, we will provide an overview of these systems.

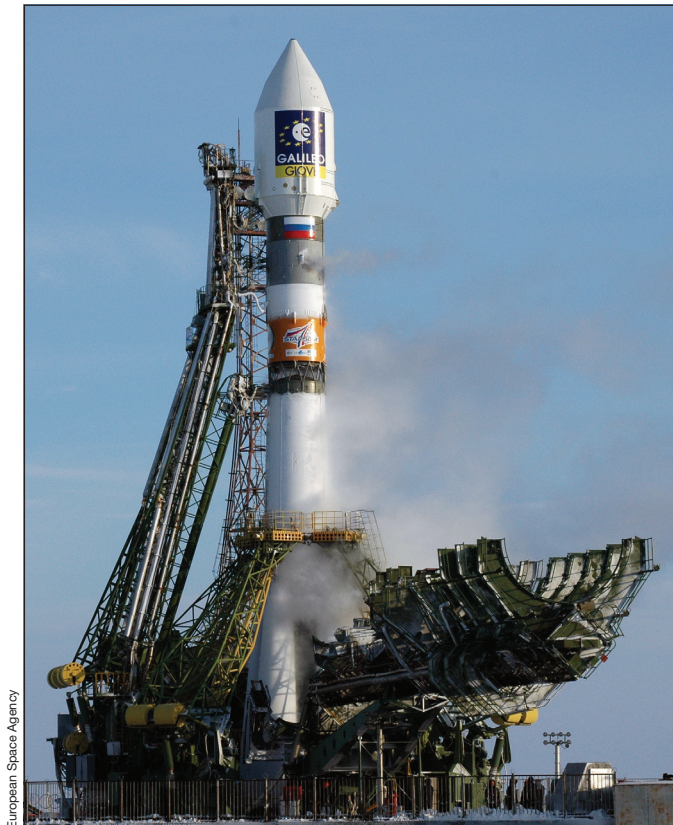


Figure 23 Launch of Galileo Satellite

Currently, the following GNSS systems are operational:

- GPS (United States)
- GLONASS (Russia)

The following GNSS systems are planned and are in varying stages of development:

- Galileo (European Union)
- Compass (China)

The following regional navigation satellite systems are planned and are in varying stages of development:

- IRNSS (India)
- QZSS (Japan)

GPS (Global Positioning System, United States)

GPS was the first GNSS system. GPS (or NAVSTAR, as it is officially called) satellites were first launched in the late 1970's and early 1980's for the US Department of Defense. Since that time, several generations (referred to as "Blocks") of GPS satellites have been launched. Initially, GPS was available only for military use but in 1983, a decision was made to extend GPS to civilian use. A GPS satellite is depicted in Figure 24.



Figure 24 GPS IIR-M Satellite (artist's rendition)

Space Segment

The GPS space segment is summarized in Table 2. The orbit period of each satellite is approximately 12 hours, so this provides a GPS receiver with at least six satellites in view from any point on Earth, under open-sky conditions.

Satellites	21 plus 3 spares
Orbital planes	6
Orbit inclination	55 degrees
Orbit radius	26,560 km

Table 2 GPS Satellite Constellation

A GPS satellite orbit is illustrated in Figure 25.

GPS satellites continually broadcast their identification, ranging signals, satellite status and corrected ephemerides (orbit parameters). The satellites are identified either by their Space Vehicle Number (SVN) or their PseudoRandom code Number (PRN).

Signals

Table 3 provides further information on GPS signals. GPS signals are based on CDMA (Code Division Multiple Access) technology, which we discussed in Chapter 2.

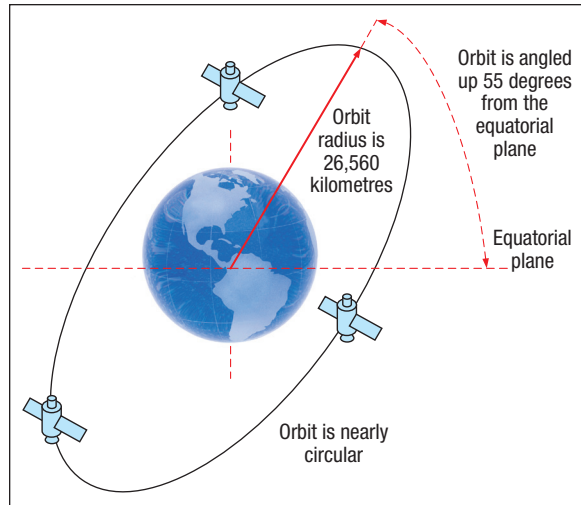


Figure 25 GPS Satellite Orbit

Designation	Frequency	Description
L1	1575.42 MHz	L1 is modulated by the C/A code (Coarse/Acquisition) and the P-code (Precision) which is encrypted for military and other authorized users.
L2	1227.60 MHz	L2 is modulated by the P-code and, beginning with the Block IIR-M satellites, the L2C (civilian) code. L2C, which is considered “under development”, is discussed below, under “GPS Modernization”.
L5	1176.45 MHz	At the time of writing, L5 is available for demonstration on one GPS satellite. The L5 signal is discussed below, under “GPS Modernization”.

Table 3 GPS Signal Characteristics

Control Segment

The GPS control segment consists of a master control station (and a backup master control station) and monitor stations throughout the world, as shown in Figure 26.



Figure 26 GPS Control Segment

Four monitor stations were implemented early in the NAVSTAR program, then six more NGA (National Geospatial Intelligence Agency, also part of the United States Department of Defense) stations were added in 2005.

The monitor stations track the satellites via their broadcast signals, which contain satellite ephemeris data, ranging signals, clock data and almanac data. These signals are passed to the master control station where the ephemerides are recalculated. The resulting ephemeride and timing corrections are transmitted back up to the satellites through data up-loading stations.

GPS Modernization

GPS reached Fully Operational Capability in 1995. In 2000, a project was initiated to modernize the GPS space and ground segments, to take advantage of new technologies and user requirements.

Space segment modernization has included new signals, as well as improvements in atomic clock accuracy, satellite signal strength and reliability. Control segment modernization includes improved ionospheric and tropospheric modelling and in-orbit accuracy, and additional monitoring stations. User equipment has also evolved, to take advantage of space and control segment improvements.

L2C

The latest generation of GPS satellites has the capability to transmit a new civilian signal, designated L2C. Once operational, L2C will ensure the accessibility of two civilian codes. L2C will be easier for the user segment to track and it will provide improved navigation accuracy. It will also provide the ability to directly measure and remove the ionospheric delay error for a particular satellite, using the civilian signals on both L1 and L2.

L5 (Safety of Life)

United States has started implementing a third civil GPS frequency (L5) at 1176.45 MHz. The first NAVSTAR GPS satellite to transmit L5, on a demonstration basis, was launched in 2009.

The benefits of the L5 signal include meeting the requirements for critical safety-of-life applications such as that needed for civil aviation, and providing:

- Improved ionospheric correction.
- Signal redundancy.
- Improved signal accuracy.
- Improved interference rejection.

Other

In addition to the new L2C and L5 signals, GPS satellite modernization includes a new military signal and an improved L1C which will be backwards compatible with L1 and which will provide greater civilian interoperability with Galileo.

GLONASS (Global Navigation Satellite System, Russia)

GLONASS was developed by the Soviet Union as an experimental military communications system during the 1970s. When the Cold War ended, the Soviet Union recognized that GLONASS had commercial applications, through the system's ability to transmit weather broadcasts, communications, navigation and reconnaissance data.

The first GLONASS satellite was launched in 1982 and the system was declared fully operational in 1993. After a period where GLONASS performance declined, Russia committed to bringing the system up to the required minimum of 18 active satellites. The Russian government set 2011 as the date for full deployment of the 24-satellite constellation and has ensured that the necessary financial support will be there to meet this date.

GLONASS satellites have evolved since the first ones were launched. The latest generation, GLONASS-M, is shown being readied for launch in Figure 27.

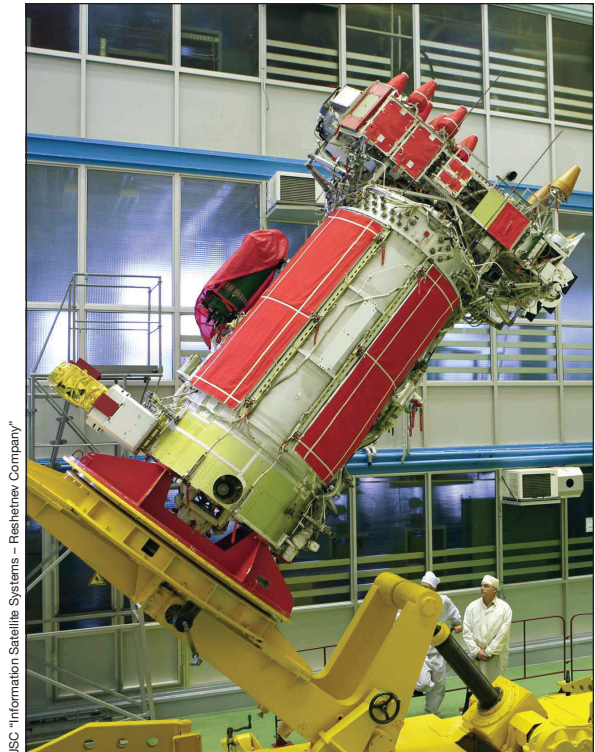


Figure 27 GLONASS-M Satellite in Final Manufacturing

GLONASS System Design

The GLONASS constellation provides visibility to a variable number of satellites, depending on your location. A minimum of four satellites in view allows a GLONASS receiver to compute its position in three dimensions and to synchronize with system time.

GLONASS Space Segment

The GLONASS space segment is summarized in Table 4.

When complete, the GLONASS space segment will consist of 24 satellites in three orbital planes, with eight satellites per plane.

The GLONASS constellation geometry repeats about once every eight days. The orbit period of each satellite is approximately 8/17 of a sidereal¹ day so that, after eight sidereal days, the GLONASS satellites have completed exactly 17 orbital revolutions.

Each orbital plane contains eight equally spaced satellites. One of the satellites will be at the same spot in the sky at the same sidereal time each day.

The satellites are placed into nominally circular orbits with target inclinations of 64.8 degrees and an orbital radius of 25,510 km, about 1,050 km lower than GPS satellites.

The GLONASS satellite signal identifies the satellite and includes:

- Positioning, velocity and acceleration information for computing satellite locations.
- Satellite health information.
- Offset of GLONASS time from UTC (SU) [formerly Soviet Union and now Russia].
- Almanac of all other GLONASS satellites.

*“The Earth was absolutely round . . . I never knew what the word ‘round’ meant until I saw Earth from space.”
Alexei Leonov, Soviet astronaut, talking about his historic 1985 spacewalk.*

Satellites	21 plus 3 spares
Orbital planes	3
Orbital inclination	64.8 degrees
Orbit radius	25,510 km

Table 4 GLONASS Satellite Constellation



Figure 28 View of Earth (as seen by Apollo 17 crew)

¹ A sidereal day is the time for one complete rotation of the earth relative to a particular star. A sidereal day is about four minutes shorter than a mean solar day.

GLONASS Control Segment

The GLONASS control segment consists of the system control center and a network of command tracking stations across Russia. The GLONASS control segment, similar to that of GPS, monitors the status of satellites, determines the ephemeride corrections, and satellite clock offsets with respect to GLONASS time and UTC (Coordinated Universal Time). Twice a day, it uploads corrections to the satellites.

GLONASS Signals

Table 5 summarizes the GLONASS signals.

Designation	Frequency	Description
L1	1598.0625 - 1609.3125 MHz	L1 is modulated by the HP (high precision) and the SP (standard precision) signals.
L2	1242.9375 - 1251.6875 MHz	L2 is modulated by the HP and SP signals. The SP code is identical to that transmitted on L1.

Table 5 GLONASS Signal Characteristics

GLONASS satellites each transmit on slightly different L1 and L2 frequencies, with the P-code (HP code) on both L1 and L2, and the C/A code (SP code), on L1 (all satellites) and L2 (most satellites). GLONASS satellites transmit the same code at different frequencies, a technique known as FDMA, for frequency division multiple access. Note that this is a different technique from that used by GPS.

GLONASS signals have the same polarization (orientation of the electromagnetic waves) as GPS signals, and have comparable signal strength.

The GLONASS system is based on 24 satellites using 12 frequencies. It achieves this by having antipodal satellites transmitting on the same frequency. Antipodal satellites are in the same orbital plane but are separated by 180 degrees. The paired satellites can transmit on the same frequency because they will never appear at the same time in view of a receiver on the Earth's surface, as shown in Figure 29.

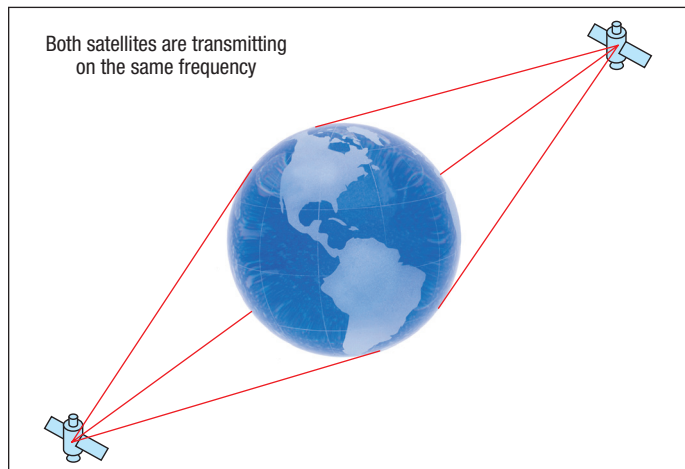


Figure 29 GLONASS Antipodal Satellites

GLONASS and GPS

The use of GLONASS in addition to GPS, results in there being a larger number of satellites in the field of view, which has the following benefits:

- Reduced signal acquisition time
- Improved position and time accuracy
- Reduction of problems caused by obstructions such as buildings and foliage
- Improved spatial distribution of visible satellites, resulting in improved dilution of precision.

We will say more about dilution of precision in Chapter 4.

To determine a position in GPS-only mode, a receiver must track a minimum of four satellites. In combined GPS/GLONASS mode, the receiver must track five satellites, at least one of which must be a GLONASS satellite, so the receiver can determine the GPS/GLONASS time offset.

With the availability of combined GPS/GLONASS receivers, users have access to a satellite-combined system with over 40 satellites. Performance in urban canyons and other locations with restricted visibility improves as more satellites are accessible by the receiver.

Galileo (European Union)

*“Measure what is measurable, and make measurable what is not so.”
Galilei Galileo, Italian physicist, mathematician, astronomer, and philosopher.*

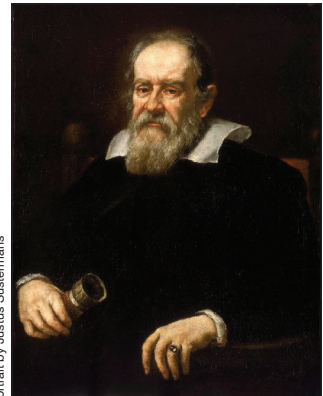
In May 1999, a mountaineering expedition carried a GPS receiver to the summit of Mount Everest, allowing them to measure its elevation at 8,850 m (29,035 ft). We think Galileo would have been happy.

Galileo, Europe’s planned global navigation satellite system, will provide a highly accurate and guaranteed global positioning service under civilian control. The United States and European Union have been cooperating since 2004 to ensure that GPS and Galileo are compatible and interoperable at the user level.

By offering dual frequencies as standard, Galileo will deliver real-time positioning accuracy down to the metre range, previously not achievable by a publicly available system.

Galileo will guarantee availability of service under all but the most extreme circumstances and it will inform users within seconds of a failure of any satellite. This will make it suitable for applications where safety is crucial, such as in air and ground transportation.

The first experimental Galileo satellite, part of the Galileo System Test Bed (GSTB) was launched in December 2005. The purpose of this experimental satellite is to characterize critical Galileo technologies, which are already in development under European Space Agency (ESA) contracts. Four operational satellites are planned to be launched in the 2010-2012 time frame to validate the basic



portrait by Justus Sustermans

Figure 30 Galilei Galileo

Galileo space and ground segment. Once this In-Orbit Validation (IOV) phase has been completed, the remaining satellites will be launched, with plans to reach Full Operational Capability (FOC) likely sometime after 2013.

System Design

The Galileo space segment is summarized in Table 6. Once the constellation is operational, Galileo navigation signals will provide coverage at all latitudes. The large number of satellites, together with the optimization of the constellation and the availability of the three active spare satellites, will ensure that the loss of one satellite has no discernable effect on the user segment.

Satellites	27 operational and three active spares
Orbital planes	3
Orbital inclination	56 degrees
Orbit radius	23,616 km

Table 6 Galileo Satellite Constellation

Two Galileo Control Centres (GCC), to be located in Europe, will control the satellites. Data recovered by a global network of twenty Galileo Sensor Stations (GSS) will be sent to the GCC through a redundant communications network. The GCCs will use the data from the Sensor Stations to compute integrity information and to synchronize satellite time with ground station clocks. Control Centres will communicate with the satellites through uplink stations, which will be installed around the world.

Galileo will provide a global Search and Rescue (SAR) function, based on the operational search and rescue satellite-aided Cospas-Sarsat² system. To do this, each Galileo satellite will be equipped with a transponder that will transfer distress signals to the Rescue Coordination Centre (RCC), which will then initiate the rescue operation. At the same time, the system will provide a signal to the user, informing them that their situation has been detected and that help is underway. This latter feature is new and is considered a major upgrade over existing systems, which do not provide feedback to the user.



Figure 31 Galileo Satellite in Orbit

² Cospas-Sarsat is an international satellite-based search and rescue (SAR) distress alert detection and information distribution system, established by Canada, France, United States, and the former Soviet Union in 1979.

Five Galileo services are proposed, as summarized in Table 7.

Service	Description
Free Open Service (OS)	Provides positioning, navigation and precise timing service. It will be available for use by any person with a Galileo receiver. No authorisation will be required to access this service. Galileo is expected to be similar to GPS in this respect.
Highly reliable Commercial Service (CS)	Service providers can provide added-value services, for which they can charge the end customer. The CS signal will contain data relating to these additional commercial services.
Safety-of-Life Service (SOL)	Improves on OS by providing timely warnings to users when it fails to meet certain margins of accuracy. A service guarantee will likely be provided for this service.
Government encrypted Public Regulated Service (PRS)	Highly encrypted restricted-access service offered to government agencies that require a high availability navigation signal.
Search and Rescue Service (SAR)	Public service designed to support search and rescue operations, which will make it possible to locate people and vehicles in distress.

Table 7 Galileo Services

Compass (China)

China has started the implementation of a GNSS system, known as Compass or Beidou-2. The initial system will provide regional coverage. The target is that this be followed after 2015 with the implementation of a constellation of GEO (geostationary orbit) and MEO satellites that will provide global coverage, as shown in Table 8.

Satellites	35, a combination of 5 GEO and 30 MEO
Orbital planes	6
Orbital inclination	55 degrees
Orbit radius	27,528 km

Table 8 Planned Compass Satellite Constellation

Two levels of service will be provided:

- Public service, for civilian use, and free to users in China. The public service will provide location accuracy of 10 metres, velocity accuracy within 0.2 metres per second and timing accuracy of 50 nanoseconds.
- Licensed military service, more accurate than the public service, and also providing system status information and military communications capability.



IRNSS (Indian Regional Navigation Satellite System, India)

India plans to launch its own regional navigation satellite system to provide regional coverage in the 2011-2012 timeframe. The IRNSS system will consist of seven satellites, three of them in geosynchronous orbits, the other four in geostationary orbits.³ The system will provide a position accuracy of better than 20 metres throughout India.

QZSS (Quasi-Zenith Satellite System, Japan)

Japan plans to launch demonstration satellites for the QZSS, a three-satellite system that will provide regional communication services and positioning information for the mobile environment. QZSS will provide limited accuracy in standalone mode, so it is viewed as a GNSS augmentation service. QZSS satellites will be placed in a periodic Highly Elliptical Orbit (HEO). These orbits will allow the satellites to “dwell” for more than 12 hours a day at an elevation above 70° (meaning they appear almost overhead most of the time).

Closing Remarks

“By perseverance the snail reached the ark.” Charles Haddon Spurgeon, English preacher.

Now that you know a bit more about global navigation satellite systems, we will discuss advanced GNSS concepts in Chapter 4 and GNSS applications and equipment in Chapter 5.

³ A geosynchronous orbit has an orbital period matching the Earth’s sidereal rotation period. This synchronization means that for an observer at a fixed location on Earth, a satellite in a geosynchronous orbit returns to exactly the same place in the sky at exactly the same time each day. The term geostationary is used to refer to the special case of a geosynchronous orbit that is circular (or nearly circular) and at zero (or nearly zero) inclination, that is, directly above the equator. Satellites in geostationary orbits appear stationary at one location at all times.



Chapter 4

Advanced GNSS Concepts

In Chapter 2, we introduced the basic concepts of GNSS positioning, specifically as they apply to single-point positioning, where a single GNSS receiver operates individually, or “standalone,” to determine its location and time. In this chapter, we introduce methods by which GNSS receivers improve performance by using more advanced techniques.

Differential GNSS

A commonly used technique for improving GNSS performance is differential GNSS, which is illustrated in Figure 32.

Using differential GNSS, the position of a fixed GNSS receiver, referred to as a “base station,” is determined to a high degree of accuracy using conventional surveying techniques. The base station determines ranges to GNSS satellites in view using two methods:

- Using the code-based positioning technique described in Chapter 2.
- Using the (precisely) known locations of the base station and the satellites, the location of satellites being determined from the precisely known orbit ephemerides and satellite time.

The base station compares the ranges. Differences between the ranges can be attributed to satellite ephemeris and clock errors, but mostly to errors associated with atmospheric delay. Base stations send these errors to other receivers (rovers), which incorporate the corrections into their position calculations.

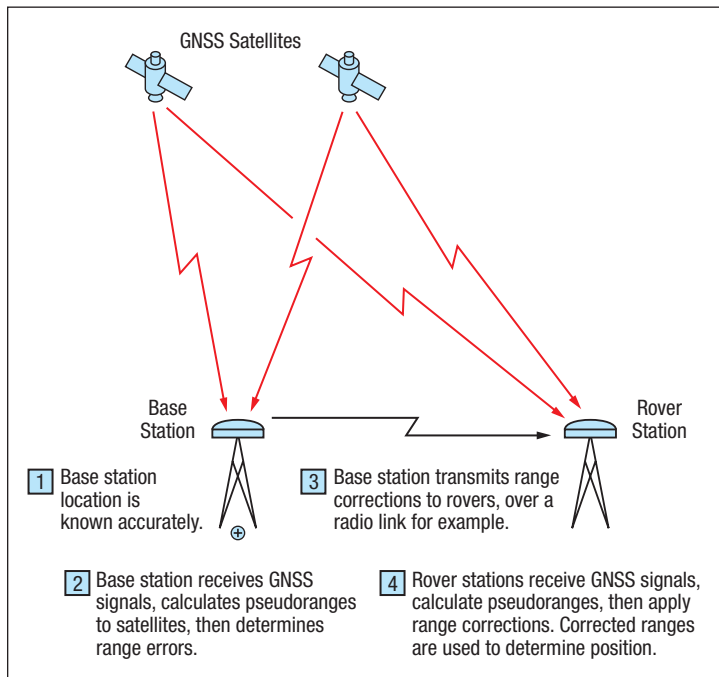


Figure 32 Differential GNSS

Differential positioning requires a data link between base stations and rovers if corrections need to be applied in real-time, and at least four GNSS satellites in view at both the base station and the rovers. The absolute accuracy of the rover's computed position will depend on the absolute accuracy of the base station's position.

Since GNSS satellites orbit high above the earth, the propagation paths from the satellites to the base stations and rovers pass through similar atmospheric conditions, as long as the base station and rovers are not too far apart. Differential GNSS works very well with base-station-to-rover separations of up to tens of kilometres.

Satellite-based Augmentation Systems

For applications where the cost of a differential GNSS system is not justified, or if the rover stations are spread over too large an area, a Satellite-Based Augmentation System (SBAS) may be more appropriate for enhancing position accuracy.

SBAS systems are geosynchronous¹ satellite systems that provide services for improving the accuracy, integrity, and availability of basic GNSS signals.

- Accuracy is enhanced through the transmission of wide-area corrections for GNSS range errors.
- Integrity is enhanced by the SBAS network quickly detecting satellite signal errors and sending alerts to receivers that they should not track the failed satellite.
- Signal availability can be improved if the SBAS transmits ranging signals from its satellites.

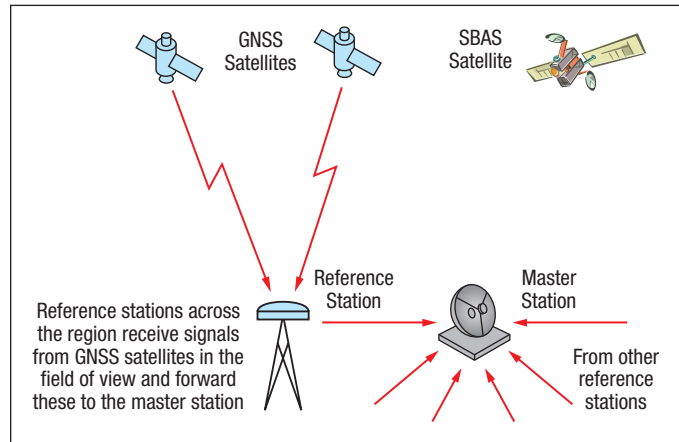


Figure 33 SBAS – GNSS Data Gathering

SBAS systems include reference stations, master stations, uplink stations and geosynchronous satellites, as shown in Figure 33.

Reference stations, which are geographically distributed throughout the SBAS service area, receive GNSS signals and forward them to the master station. Since the locations of the reference stations are accurately known, the master station can accurately calculate wide-area corrections.

¹ As mentioned in Chapter 3, a geosynchronous orbit has an orbital period matching the Earth's sidereal rotation period. This synchronization means that for an observer at a fixed location on Earth, a satellite in a geosynchronous orbit returns to exactly the same place in the sky at exactly the same time each day. The term geostationary is used to refer to the special case of a geosynchronous orbit that is circular (or nearly circular) and at zero (or nearly zero) inclination, that is, directly above the equator. Satellites in geostationary orbits appear stationary at one location at all times.

As shown in Figure 34, corrections are uplinked to the SBAS satellite then broadcast to GNSS receivers throughout the SBAS coverage area, as shown in Figure 35.

User equipment receives the corrections and applies them to range calculations.

SBAS services can be grouped into two categories:

- Free SBAS Services
- Commercial SBAS Services

In general, free government-provided SBAS services use the same frequency as GPS (CDGPS is an exception), and commercial SBAS services use a different frequency. In this case, additional equipment may be required.

Free SBAS Services

The following sections provide an overview of some of the free SBAS services that have been implemented around the world or that are planned.

Wide Area Augmentation System (WAAS)

The US Federal Aviation Administration (FAA) has developed the Wide Area Augmentation System (WAAS) to provide GPS corrections and a certified

level of integrity to the aviation industry, to enable aircraft to conduct varying levels of precision approach to airports. The corrections are also available free of charge to civilian users in North America.

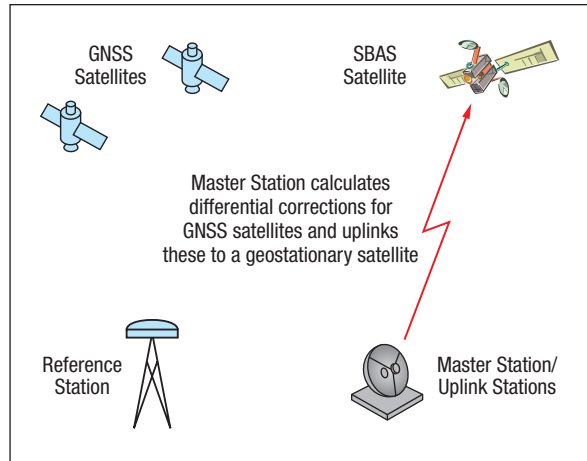


Figure 34 SBAS – Correction Calculation and Uplink

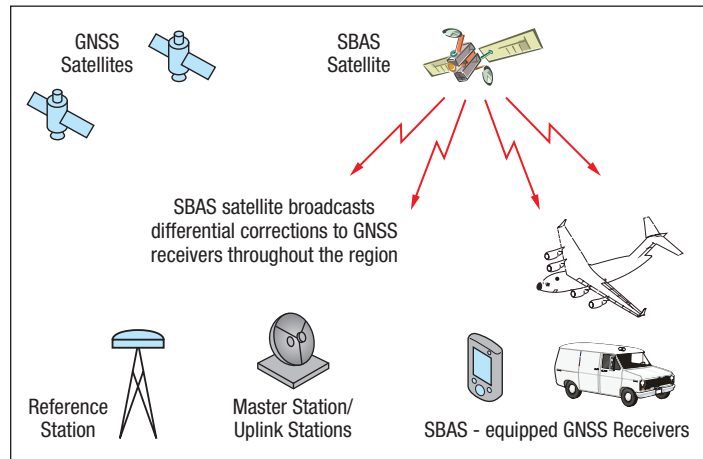


Figure 35 SBAS – Correction Broadcast

A Wide Area Master Station (WMS) receives GPS data from Wide Area Reference Stations (WRS) located throughout the United States. The WMS calculates differential corrections then uplinks these to two WAAS geostationary satellites for broadcast across the United States, as shown in Figure 36.

Separate corrections are calculated for ionospheric delay, satellite timing, and satellite orbits, which allows error corrections to be processed separately, if appropriate, by the user application.

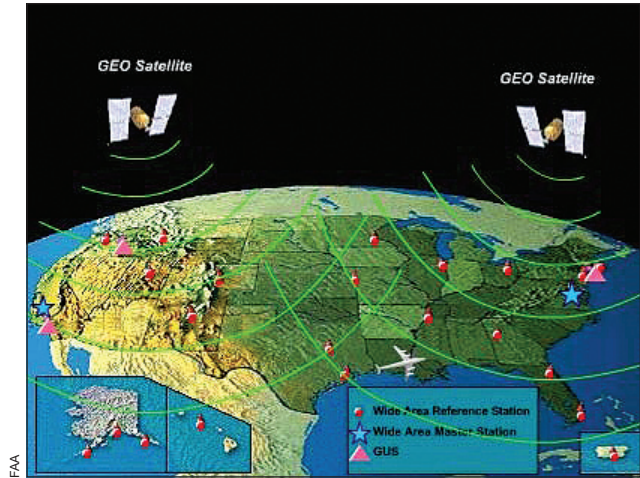


Figure 36 Wide-Area Augmentation System (WAAS)

WAAS broadcasts correction data on the same frequency as GPS, which allows for the use of the same receiver and antenna equipment as that used for GPS. To receive correction data, user equipment must have line of sight to one of the WAAS satellites.

European Geostationary Navigation Overlay Service (EGNOS)

The European Space Agency, in cooperation with the European Commission (EC) and EUROCONTROL (European Organization for the Safety of Air Navigation) has developed the European Geostationary Navigation Overlay Service (EGNOS), a regional augmentation system that improves the accuracy of positions derived from GPS signals and alerts users about the reliability of the GPS signals.

Two EGNOS satellites cover European Union member nations and several other countries in Europe. EGNOS is expected to be certified for safety-of-life applications in 2010. It transmits differential correction data for public use. EGNOS satellites have also been placed over the eastern Atlantic Ocean, the Indian Ocean, and the African mid-continent.

CDGPS

CDGPS is currently offered throughout Canada, most of the continental United States, and much of the Arctic. CDGPS corrections are broadcast over Canada's

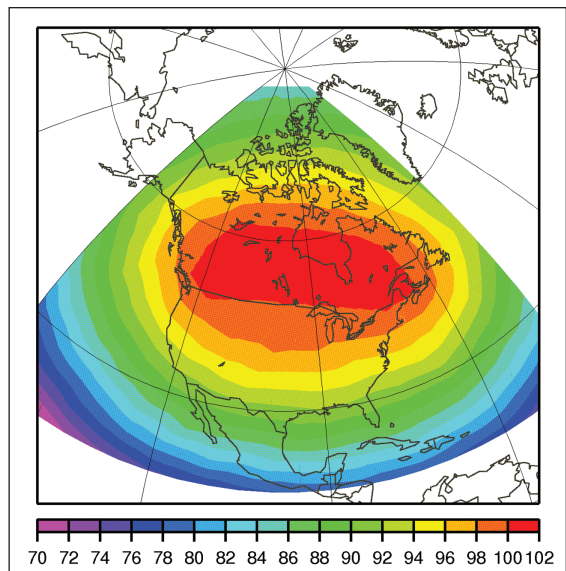


Figure 37 CDGPS Coverage

MSAT-1 satellite, as shown in Figure 37. Unlike other SBAS services, CDGPS is broadcast on slightly different L-band frequencies than GPS, so additional equipment, or receivers designed for CDGPS are required.

Other SBAS Systems

In Japan, the MTSAT Satellite Based Augmentation System (MSAS) is being developed by the Japan Civil Aviation Bureau (JCAB).

India is planning to implement GAGAN (GPS-Aided GEO Augmented Navigation system), a regional SBAS that will support flight navigation over Indian airspace by 2011. The system will be based on an Indian geostationary satellite, eight reference stations installed throughout India, and a master control centre at Bangalore. GAGAN will be compatible with other SBAS systems, such as WAAS, EGNOS and MSAS.

China is planning SNAS (Satellite Navigation Augmentation System), to provide WAAS-like service for the China region.

Commercial SBAS Services

There are several providers of fee-based commercial SBAS service, including OmniSTAR, VERIPOS, which provides differential GPS corrections to the offshore oil and gas industry, and StarFire, a wide-area DGPS system developed by John Deere. To illustrate commercial SBAS service, the following section provides more detail on the OmniSTAR service.

OmniSTAR

As mentioned previously, OmniSTAR is a subscription-based service that transmits differential corrections at L-band frequencies close to GPS frequencies, allowing for a single antenna to be used.

OmniSTAR provides three levels of service – VBS, providing sub-metre horizontal accuracy, XP, providing decimetre (tenth of a metre) accuracy, and HP, providing sub-decimetre accuracy.

OmniSTAR satellites provide coverage over most of the world's land areas. Satellite footprints have been tailored to provide coverage of a continental area or geographic region. Differential corrections from all base stations in the region are broadcast every few seconds.

Real-Time Kinematic (RTK)

The positioning technique we described in Chapter 2 is referred to as code-based positioning, because the receiver correlates with and uses the pseudorandom codes transmitted by four or more satellites to determine the ranges to the satellites. From these ranges and knowing where the satellites are, the receiver can establish its position to within a few metres.

For applications such as surveying, higher accuracies are required. Real-Time Kinematic (RTK), a technique that uses carrier-based ranging, provides ranges (and therefore positions) that are orders of magnitude more precise than those available through code-based positioning.

Real-time kinematic techniques are complicated. The basic concept is to reduce and remove errors common to a base station and rover pair, as illustrated in Figure 38. These error sources were discussed in Chapter 2.

At a very basic conceptual level, as shown in Figure 38, the range is calculated by determining the number of carrier cycles between the satellite and the rover station, then multiplying this number by the carrier wavelength.

The calculated ranges still include errors from such sources as satellite clock and ephemerides, and ionospheric and tropospheric delays. To eliminate these errors and to take advantage

of the precision of carrier-based measurements, RTK performance requires measurements to be transmitted from the base station to the rover station.

A complicated process called “ambiguity resolution” is needed to determine the number of whole cycles. For further information about ambiguity resolution, see the references at the back of this book.

Rovers determine their position using algorithms that incorporate ambiguity resolution and differential correction. Like DGNS, the position accuracy achievable by the rover depends on, among other things, its distance from the base station (referred to as the “baseline”) and the accuracy of the differential corrections. Corrections are as accurate as the known location of the base station and the quality of the base station’s satellite observations. Site selection is important for minimizing environmental effects such as interference and multipath, as is the quality of the base station and rover receivers and antennas.

Virtual Reference Stations (VRS)

VRS is based on the use of several widely spaced permanent stations. Depending on the implementation, positioning data from the permanent stations is regularly communicated to a central processing station. On demand from RTK user terminals, which transmit their approximate location to the central station, the central station calculates and transmits correction information or corrected position to the RTK user terminal. The benefit of this approach is an overall reduction in the number of RTK base stations required. Depending on the implementation, data may be transmitted over cellular radio links or other wireless medium.

Dilution of Precision (DOP)

The geometric arrangement of satellites, as they are presented to the receiver, affects the accuracy of position and time calculations. Receivers will ideally be designed to use signals from available satellites in a manner that minimizes this so called “dilution of precision.”

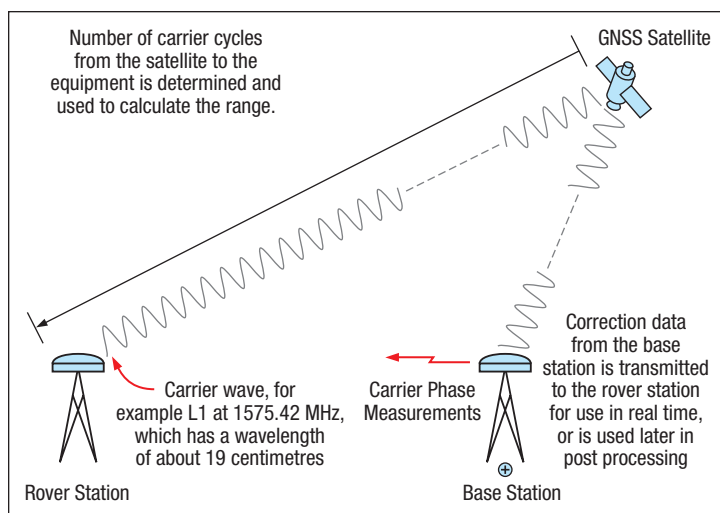


Figure 38 Real-Time Kinematic

To illustrate DOP, consider the example shown in Figure 39, where the satellites being tracked are clustered in a small region of the sky.

In the example shown in Figure 39, intentionally a bit extreme to illustrate the effect of DOP, it is difficult to determine where the ranges intersect. Position is “spread” over the area of range intersections, an area which is enlarged by range inaccuracies (which can be viewed as a “thickening” of the range lines).

As shown in Figure 40 below, the addition of a range measurement to a satellite that is angularly separated from the cluster allows us to determine a fix more precisely.

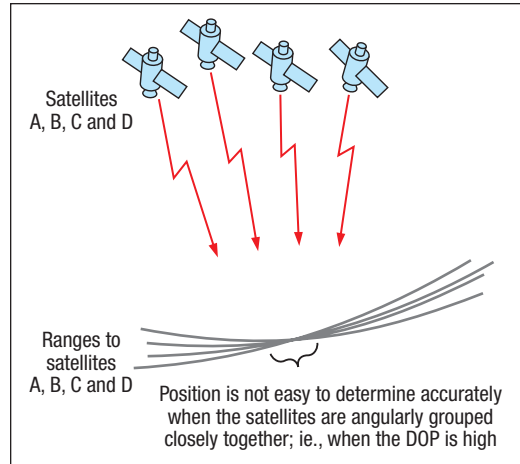


Figure 39 Dilution of Precision (poor satellite geometry)

Although it is calculated using complex statistical methods, we can say the following about DOP:

- DOP is a numerical representation of satellite geometry, and it is dependent on the locations of satellites that are visible to the receiver.
- The smaller the value of DOP, the more precise the result of the time or position calculation. The relationship is shown in the following formula:

$$\text{Inaccuracy of Position Measurement} = \text{DOP} \times \text{Inaccuracy of Range Measurement}$$

So, if DOP is very high, the inaccuracy of the position measurement will be much larger than the inaccuracy of the range measurement.

- DOP can be used as the basis for selecting the satellites on which the position solution will be based; specifically, selecting satellites to minimize DOP for a particular application.
- A DOP above 6 results in generally unacceptable accuracies for DGPS and RTK operations
- DOP varies with time of day and geographic location but, for a fixed position, the geometric presentation of the satellites repeats every day, for GPS.
- DOP can be calculated without determining the range. All that is needed is the satellite positions and the approximate receiver location.

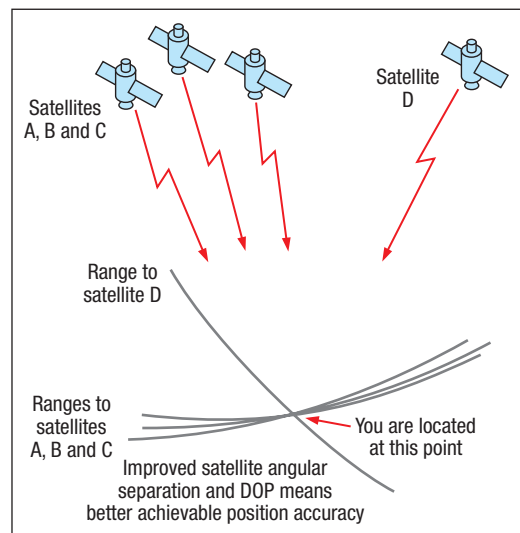


Figure 40 Dilution of Precision (improved geometry)

DOP can be expressed as a number of separate elements that define the dilution of precision for a particular type of measurement, for example, HDOP (Horizontal Dilution of Precision), VDOP (Vertical Dilution of Precision), and PDOP (Position Dilution of Precision). These factors are mathematically related. In some cases, for example when satellites are low in the sky, HDOP is low and it will therefore be possible to get a good to excellent determination of horizontal position (latitude and longitude), but VDOP may only be adequate for a moderate altitude determination. Similarly, when satellites are clustered high in the sky, VDOP is better than HDOP.

When we extend our DOP illustration to three satellites, one way to view dilution of precision is to consider the “tetrahedron” formed by having the satellites at three corners and the receiver at the fourth, as illustrated in Figure 41.

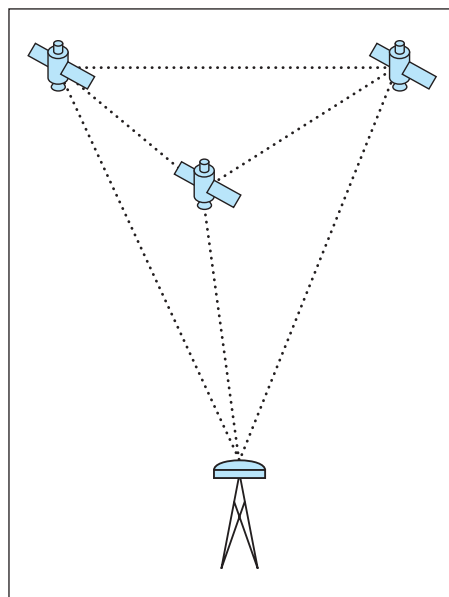


Figure 41 Minimizing Dilution of Precision

Minimizing DOP is not unlike maximizing the volume of this tetrahedron. When the satellites are tightly clustered and the angle between the satellites is small, the tetrahedron is long and narrow. The volume of the tetrahedron is small and DOP is correspondingly high (undesirable). When the satellites are all located near the horizon, the tetrahedron is flat. Again, the volume of the tetrahedron is small and DOP is high. When the satellites are not tightly clustered in the sky or low in elevation, the volume of the tetrahedron approaches a maximum and DOP is at its lowest (desirable).

In Canada and in other countries at high latitude, GNSS satellites are lower in the sky, and achieving optimal DOP for some applications, particularly where good VDOP is required, is sometimes a challenge.

When there were fewer GNSS satellites, achieving good DOP was sometimes difficult. These difficulties are being reduced with more GNSS constellations and satellites coming on line every year. Applications where the available satellites are low on the horizon or angularly clustered, such as those in urban environments or in deep open-pit mining, may still expose users to the pitfalls of DOP. If you know your application will have obstructed conditions, you may want to use a mission planning tool to determine the ideal time – the time with the ideal DOP – for your survey, as an example.

Combined GNSS/Inertial Navigation Systems

As discussed, Global Navigation Satellite Systems use signals from orbiting satellites to compute position, time and velocity. GNSS navigation has excellent accuracy provided the antenna has good visibility to the satellites. When the line of sight to satellites is blocked by obstructions such as trees or buildings, navigation becomes unreliable or impossible. Inertial Navigation Systems (INS) use

rotation and acceleration information from an Inertial Measurement Unit (IMU) to compute accurate position over time. An INS can also solve the attitude (roll, pitch and heading) of a vehicle and is not reliant on any external measurement to compute a solution. In the absence of an external reference, however, the INS solution drifts over time due to accumulating errors in the IMU data. When combined, the two techniques (GNSS and INS) enhance each other to provide a powerful navigation solution, as illustrated in Figure 42.

The degree with which the GNSS and INS technologies are integrated varies with product implementation. For example, in tightly coupled solutions, GNSS observations are used directly by the inertial solution to take advantage of available GNSS data, even when only a few satellites

are visible (for instance, to reset or adjust the position being output by the INS). Tightly coupled solutions allow feedback of the inertial solution into the GNSS receiver to improve GNSS performance, for example, signal reacquisition and convergence time.

To summarize, combining GNSS and INS technologies significantly increases opportunities for application development by overcoming the limitations of the individual technologies.

GNSS Data Post-Processing

For many applications, such as airborne survey, corrected GNSS positions are not required in real time. For these applications, raw GNSS satellite measurements are collected and stored for processing post-mission. Unlike real-time kinematic GNSS positioning, post-processing does not require real-time transmission of differential correction messages. This simplifies the hardware configuration greatly.

During post-processing, base station data can be used from one or more GNSS receivers. Multi-base processing helps preserve high accuracy over large project areas, which is a common occurrence for

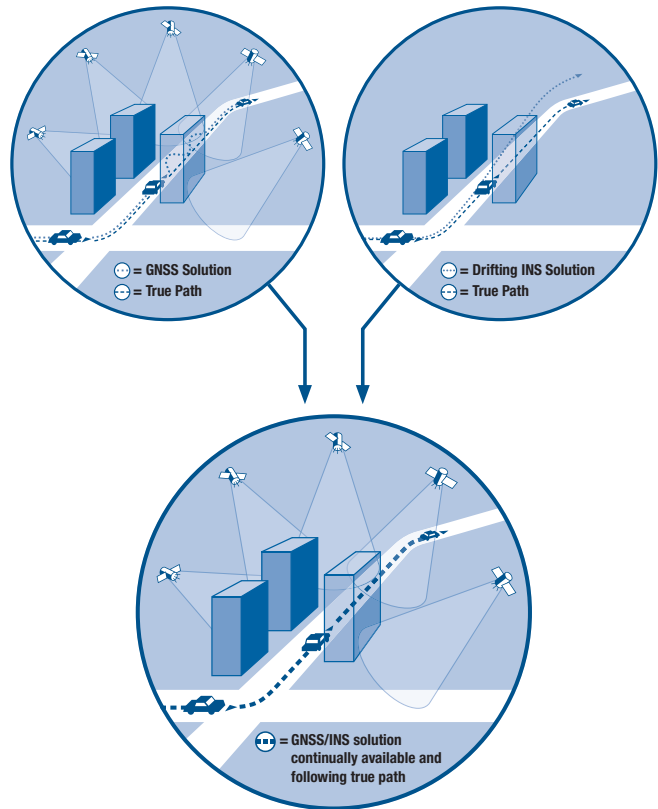


Figure 42 Combined GNSS/INS Solution

aerial applications. Depending on the project's proximity to a permanently operating GNSS network, base station data can often be freely downloaded, eliminating the need for establishing your own base station(s). Moreover, it is possible to process without any base station data through Precise Point Positioning (PPP), which utilizes downloaded precise clock and ephemeris data.

Post-processing applications offer a great deal of flexibility. Applications can involve stationary or moving base stations, and some support integration with customer or third-party software modules. Post processing applications may be designed to run on personal computers, accessible through simple-to-use graphical user interfaces, as illustrated in Figure 43.

In the example shown in Figure 43, the route taken by the vehicle is shown in the left side of the screen, and measurements recorded during the mission, such as velocity, resolved into horizontal and vertical components, are shown in the right side.

Post processing generally results in a more accurate, comprehensive solution than is possible in real time.

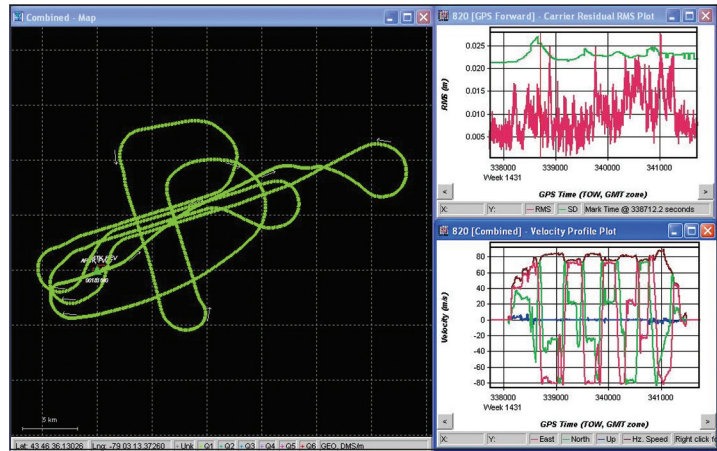


Figure 43 Post Processing of GNSS Data

Closing Remarks

This chapter has described, at a high level, some very complex GNSS concepts. If you want to learn more about these, we have provided a list of references at the end of the book.



Chapter 5

GNSS Applications and Equipment

“Every generation needs a new revolution.” Thomas Jefferson, influential Founding Father and third president of the United States.

We don’t think it is an overstatement to say that the application of GNSS is revolutionizing, and will continue to revolutionize, the way businesses and governments operate, and how we conduct our personal lives. This chapter highlights some of the incredible GNSS applications and equipment that are now available.

Applications

In a short book, it is impossible to describe all GNSS applications. While we will highlight some consumer applications of this technology, our focus will be on commercial applications in various industries, including:

- Transportation
- Machine Control
- Surveying
- Port Automation
- Timing
- Marine
- Defence

Consumer

GNSS technology has been adopted by the consumer market, in a broad range of products.

GNSS receivers are now routinely integrated into mobile communications equipment, to support applications that display maps showing the location of and best route to stores and restaurants.

Portable navigation devices can give drivers directions on road or off, as shown in Figure 44.

Geocaching is an outdoor activity in which participants use a GNSS receiver to hide and seek containers (called “geocaches” or “caches”) around the world.

Currently, most GNSS consumer products are based on GPS, but this will change as more GNSS constellations are implemented.



Figure 44 Portable Navigation Device

Transportation

“I knew I was going to take the wrong train, so I left early.” Yogi Berra.

In rail transportation, GNSS is used in conjunction with other technologies, to track the location of locomotives and rail cars, maintenance vehicles and wayside equipment, for display at central monitoring consoles. Knowing the precise location of rail equipment reduces accidents, delays, and operating costs, enhancing safety, track capacity, and customer service.

In aviation, GNSS is being used for aircraft navigation from departure, en route, to landing. GNSS facilitates aircraft navigation in remote areas that are not well served by ground-based navigation aids, and it is a significant component of collision-avoidance systems, and of systems used to improve approaches to airport runways. Refer to “Wide Area Augmentation System (WAAS)” in Chapter 4 for information about WAAS, a US system that delivers GPS corrections and a certified level of integrity to the US aviation industry, enabling aircraft to conduct varying levels of precision approach to airports.

In marine transportation, GNSS is being used to accurately determine the position of ships when they are in open sea and also when they are maneuvering in congested ports. GNSS is incorporated into underwater surveying, buoy positioning, navigation hazard location, dredging, and mapping. In a related application, GNSS has been integrated with automated container placement in large ports. We’ll talk about this application later in the chapter.



Figure 45 Automatically Guided Trenching Machine

In surface transportation, vehicle location and in-vehicle navigation systems are now being used throughout the world. Many vehicles are equipped with navigation displays that superimpose vehicle location and status on maps. GNSS is used in systems that track and forecast the movement of freight and monitor road networks, improving efficiency and enhancing driver safety.

Machine Control

GNSS technology is being integrated into equipment such as bulldozers, excavators, graders, pavers and farm machinery to enhance productivity in the real-time operation of this equipment, and to provide situational awareness information to the equipment operator. The adoption of GNSS-based machine control is similar in its impact to the earlier adoption of hydraulics technology in machinery, which has had a profound effect on productivity and reliability.

The three main areas of machine control are agriculture, construction and surface mining. Some of the benefits of GNSS-based machine control are summarized below:

- **Efficiency:** By helping the equipment operator get to the desired grade more quickly, GNSS helps speed up the work, reducing capital and operating costs.
- **Accuracy:** The precision achievable by GNSS-based solutions minimizes the need to stop work while a survey crew measures the grade.
- **Job Management:** Managers and contractors have access to accurate information about the jobsite, and the information can be viewed remotely.
- **Data Management:** Users can print out status reports, save important data and transfer files to head office.
- **Theft Detection:** GNSS allows users to define a “virtual fence” about their equipment and property, for the purpose of automatically raising an alarm when equipment is removed, then providing equipment tracking information to the authorities.

Agriculture

“Farming looks mighty easy when your plough is a pencil, and you’re a thousand miles from the corn field.” Dwight D. Eisenhower, thirty-fourth U.S. president.

Many companies have recognized the significant challenges of farming and have developed sophisticated GNSS applications to make farming easier and more efficient.

In precision agriculture, GNSS-based applications are used to support farm planning, field mapping, soil sampling, tractor guidance, and crop assessment. More precise application of fertilizers, pesticides and herbicides reduces cost and environmental impact. GNSS applications can automatically guide farm implements along the contours of the earth in a manner that controls erosion and maximizes the effectiveness of irrigation systems. Farm machinery can be operated at higher speeds, day and night, with increased accuracy. This increased accuracy saves time and fuel, and maximizes the efficiency of the operation. Operator safety is also increased by greatly reducing fatigue.



Figure 46 Precisely Aligned Crop Rows Planted by GNSS-guided Machinery

Construction

GNSS information can be used to position the cutting edge of a blade (on a bulldozer or grader, for example) or a bucket (excavator), and to compare this position against a 3D digital design to compute cut/fill amounts. “Indicate systems” provide the operator with visual cut/fill information, via a display or light bar, and the operator manually moves the machine’s blade or bucket to get to grade. Automatic systems for bulldozers/graders use the cut/fill information to drive the hydraulic controls of the machine to automatically move the machine’s blade to grade. Use of 3D machine control dramatically reduces the number of survey stakes required on a job site (each stake costs time and money). Productivity studies have repeatedly shown that the use of 3D machine control results in work being completed faster, more accurately and with significantly less rework than conventional construction methods.

Surface Mining

GNSS information is being used to efficiently manage the mining of an ore body and the movement of waste material. GNSS equipment installed on shovels and haul trucks provides position information to a computer-controlled dispatch system to optimally route haul trucks to and from each shovel. Position information is also used to track each bucket of material extracted by the shovel, to ensure that it is routed to the appropriate location in the mine (crusher, waste dump, leach pad). Position information is used by blast hole drills to improve fracturization of the rock material and control the depth of each hole that is drilled, to keep the benches level. Multi-constellation GNSS is particularly advantageous in a surface mining environment due to the obstructions caused by the mine’s walls.

Surveying

GNSS-based surveying reduces the amount of equipment and labour required to determine the position of points on the surface of the Earth, when compared with previous surveying techniques. Using GNSS, it is possible for a single surveyor to accomplish in one day what might have taken a survey crew of three people a week.

Determining a new survey position once required measuring distances and bearings from an existing (known) survey point to the new point. This required measurements using theodolites to measure angular differences and metal “chains” (long heavy tape measures), pulled taught to minimize sag and accurately



NovAtel Inc.

Figure 47 GNSS-based Surveying Equipment

measure distances. If the new and existing survey points were separated by a large distance, the process would involve multiple setups of the theodolite, then multiple angular and distance measurements.

Using GNSS, surveyors can now set up a DGNSS base station over an existing survey point and a DGNSS rover over the new point, then record the position measurement at the rover. This simplification shows why the surveying industry was one of the early civilian adopters of GNSS technology.

Aerial Photogrammetry

Aerial photogrammetry refers to the recording of images of the ground (photographs, for example) from an elevated position, such as an aircraft. Systems of this type are now more generally referred to as “remote sensing,” since the images can be taken from aircraft or from satellites.

In the past, images would have to be manually corrected for orientation, perspective and the height of the camera and location, and manually “stitched” together. This manual process would be based on the accurate alignment of known points in adjacent pictures.

By integrating the camera with GNSS and INS, it is now possible to automate the process, in real time or post mission, to “transfer” the location accuracy of the aircraft, determined from GNSS, to the image.



Figure 48 Aerial Image of Niagara Falls

Aerial photographs are used in online map systems such as Google Earth. Many of us have found our houses, and perhaps even our cars, through these applications.

GNSS technology has also been integrated with LIDAR (Light Detection and Ranging), an optical remote sensing technology used to measure range to distant targets.

It is possible to image a feature or object down to the wavelength, which at LIDAR frequencies is less than a millionth of a metre.

Ground Mapping

Products have been developed that take 360-degree panoramic photographs to support the presentation of geometrically correct images on a computer screen. These images are continuous and precisely positioned. GNSS and IMU data is recorded before the panoramic photographs are taken. Position and attitude data is programmed into the cameras, allowing onscreen determination of positions of objects in the photos, or measurements between objects.

Geospatial Information Systems (GIS)

A geospatial information system (GIS) captures, stores, analyzes, manages, and presents data that is linked to location. The data may consist of, for example, environmental or resource data. GIS is also used to map attributes for insurance companies, municipal planning, utility companies, and others. The positions associated with the data can be provided from a GNSS receiver. GIS applications can generate detailed contour maps from the data and present these maps in a digital form, as illustrated in Figure 49.

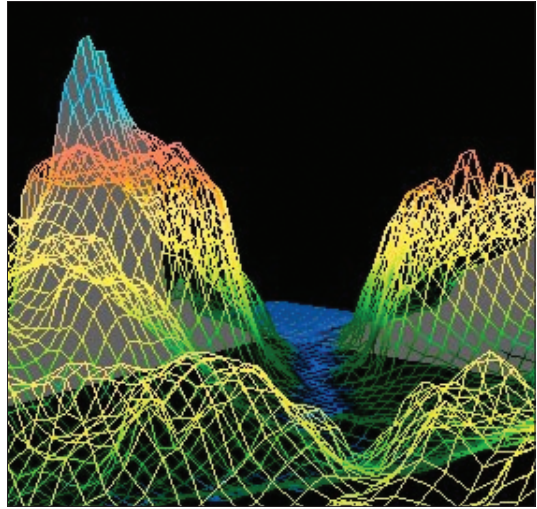


Figure 49 GIS Data Output

Port Automation

Using GNSS, shipping hubs can improve their operating efficiency by tracking the movement and placement of containers about their yards.

Gantry cranes are used in ports throughout the world to lift shipping containers, as shown in Figure 50. These cranes are large and sometimes difficult to steer accurately in a crowded shipping dock. Many cranes are equipped with GNSS-based steering devices that determine the crane's position and keep it travelling in the desired path, improving accuracy and productivity as well as the safety of operators and workers on the ground. A key benefit is the quick movement of containers about the port, which reduces food spoilage and gets toys delivered on time.



Figure 50 Gantry Crane in the Port of Vancouver

Adley Eng. NovAtel Inc.

Time Applications

“The clock, not the steam engine, is the key-machine of the modern industrial age.” Lewis Mumford, American historian of technology and science.

As we mentioned in earlier chapters, time accuracy is critical for GNSS position determination. This is why GNSS satellites are equipped with atomic clocks, accurate to nanoseconds. As part of the position determining process, the local time of GNSS receivers becomes synchronized with the very accurate satellite time. This time information, by itself, has many applications, including the synchronization of communication systems, electrical power grids, and financial networks. GNSS-derived time works well for any application where precise timing is needed by devices that are dispersed over a wide area.

Seismic monitors that are synchronized with GNSS satellite clocks can be used to determine the epicentre of an earthquake by triangulation based on the exact time the earthquake was detected by each monitor.

Marine Applications

In the Foreword, we discussed the challenges that early explorers had determining their position when at sea. With the advent of GNSS, these problems have largely disappeared.

In addition to dramatically improving marine navigation, GNSS is also being applied to a broad range of marine applications, such as oilrig positioning, underwater cable and pipeline installation and inspection, rescue and recovery, and the dredging of ports and waterways.



Figure 51 Marine Application of GNSS

GNSS-Equipped Sonobuoys

An interesting application of GNSS is the use of GNSS-equipped sonobuoys in underwater sonar systems.

Sonobuoys are dropped from aircraft over an area of interest, but are left to float autonomously. The sonobuoys detect approaching ships and other hazards in the water by transmitting sound waves through the water, detecting reflections from vessels and objects, and determining the time it takes for the “echo” to be received. The data comes up to the sonobuoy’s float then is transmitted, with GNSS positioning data, over a radio link to a survey ship. The survey ship collects and analyzes sonar data from a larger number of sonobuoys then determines and displays the location of ships and objects in the area of interest.

Defence

The defence sector makes broad use of GNSS technology, including:

- **Navigation:** Using GNSS receivers, soldiers and pilots can navigate unfamiliar terrain or conduct night-time operations. Most foot soldiers now carry hand-held GNSS receivers.
- **Search and Rescue:** If a plane crashes and that plane has a search and rescue beacon that is equipped with a GNSS receiver, it may be possible to more quickly locate it.
- **Reconnaissance and Map Creation:** The military uses GNSS to create maps of uncharted or enemy territory. They can also mark reconnaissance points using GNSS.
- **Unmanned Vehicles:** Unmanned vehicles are used primarily for reconnaissance, but they can also be used for logistics, target and decoy, and research and development.

An unmanned aerial vehicle (UAV) is an aircraft that is unoccupied but under human control, whether radio-controlled or automatically guided by a GNSS-based application. UAVs can be used to scout territory in unsecured airspace and in contaminated areas. Mission coordinates may be predefined and corrections may be sent to keep the UAV on a specific tack.

Equipment

The first generations of commercial GNSS receivers cost well over \$100,000. Now, low-cost handheld GNSS receivers cost about the same as a cellphone. Equipment vendors have developed a wide array of equipment to support the incredible range of GNSS applications that are now available. As illustrated in Figure 53, GNSS equipment consists of receivers, antennas and supporting software, in varying levels of integration and performance.



Figure 52 Predator Unmanned Aerial Vehicle (UAV)

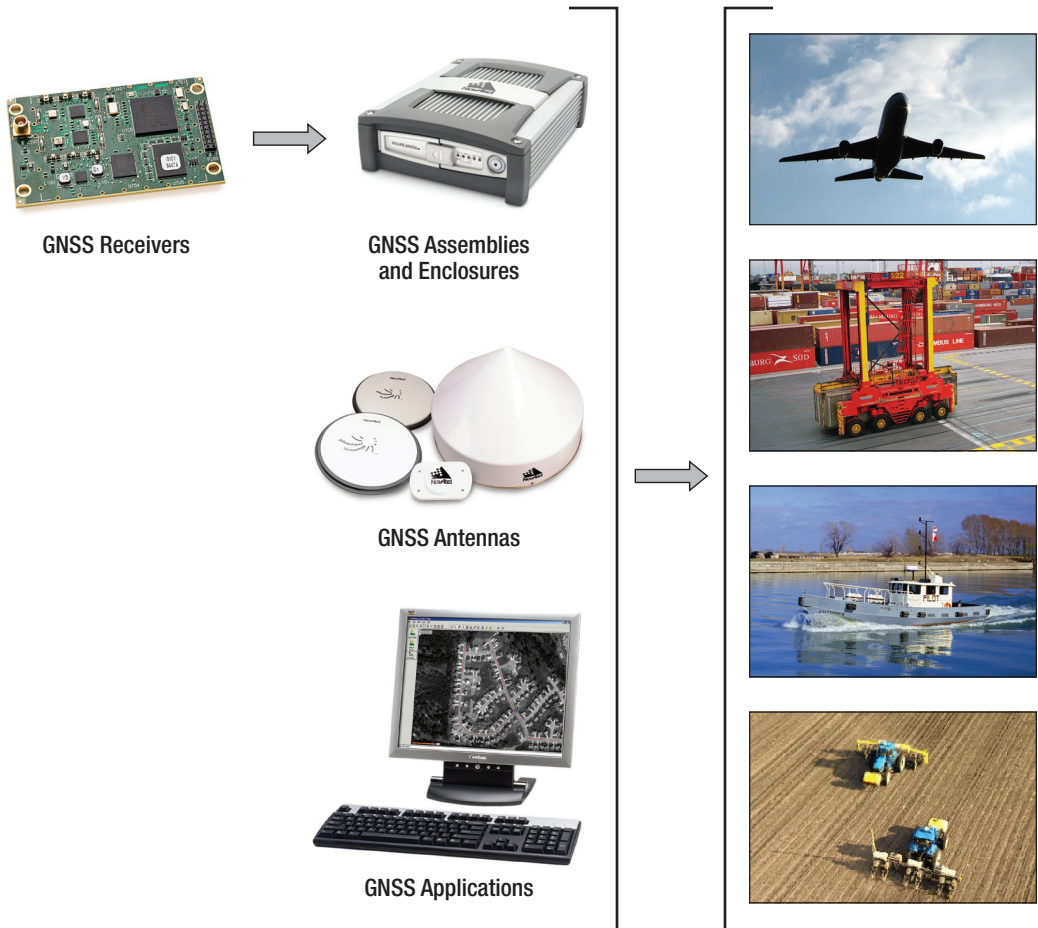


Figure 53 Examples of GNSS Equipment

Depending on the application, the antenna and receiver may be separate entities, or they may be integrated into a single package, as in a handheld GNSS receiver. GNSS equipment may be further integrated with application equipment such as a surveying instrument or a transport vessel.

GNSS equipment specifications and features depend on the application. To illustrate, users need to consider the following when selecting GNSS equipment for a particular use:

- **Accuracy:** Applications such as surveying may require centimetre-level accuracy. Others, such as positioning for hiking, may only require accuracy to within tens of metres. Some applications require absolute accuracy; that is, position defined accurately, relative to an actual reference point or location. Others may require accuracy relative to a previous position. If high-precision accuracy is obtained through the application of differential GNSS, it may be desirable that the differential service be integrated in the same package as the GNSS receiver, for example, the SBAS receiver or the radio link to the base station or rovers.

- **Acquisition Time:** For some applications, users may require a fast “time to first fix,” the time required by a GPS receiver to achieve a position solution. For other applications, it may not be important that the “fix” be available quickly. The trade-off to achieving fast acquisition time is increased probability of a wrong position/fix.
- **Reliability:** Addresses the question, “Do you need the answer (position and time) to be correct every time?”
- **Availability:** Equipment may be required to provide positioning service continuously, even in areas where signals from satellites are blocked. As we have discussed, these applications may best be served by equipment that integrates GNSS and INS equipment.

Equipment may need to support multiple constellations and frequencies, and may need to perform well in environments characterized by a high level of multipath interference. Remember, from Chapter 2, multipath interference occurs because some of the signal energy transmitted by the satellite is reflected (and therefore delayed) on the way to the receiver.

In the selection of GNSS equipment, there will almost always be tradeoffs between accuracy, acquisition time, reliability, and availability.

- **Environmental:** User equipment may have to operate over wide temperature and humidity ranges, at high altitudes, or in dusty environments. The equipment may need to be waterproof to rain or submersion.
- **Shock and Vibration:** Equipment may be subjected to high levels of shock and vibration, such as that which is characteristic of industrial vehicles.
- **Portability:** Depending on the application, the equipment may need to be hand-held (hiking applications, for example) or portable.
- **Regulatory:** Regulatory compliance will vary with the jurisdiction in which the user is operating, for example:
 - Emissions standards, such as FCC Part 15.
 - Compliance with the European Union’s Restriction of Hazardous Substances (RoHS) directive.
 - WEEE, the European Community directive that imposes responsibility for the disposal of waste electrical and electronic equipment on the equipment manufacturer.
- **Data Storage:** Receivers may be required to store time-stamped range or position information for applications that will use this information post-mission.
- **Physical Size and Power Consumption:** The user may require a receiver or antenna with a small form-factor and low power consumption for integration in a particular application.
- **User Interface:** The manner through which the user interacts with the equipment is important; for example, a keypad for entering commands, a screen for viewing position data on a map, or connectors for outputting data to other devices.

- **Computational Requirements:** Users may require that the equipment provide computed data such as velocity or heading.
- **Communications:** Position may only be useful if it is communicated to another device over, for example, a cellular radio link.
- **Future-Proof:** Although some GNSS signals and constellations may not yet be available, users may require some assurance that they will be able to use these signals and constellations once they are available.

Closing Remarks

“I used to think that cyberspace was fifty years away. What I thought was fifty years away, was only ten years away. And what I thought was ten years away... it was already here. I just wasn’t aware of it yet.”
 Bruce Sterling, American science fiction author.

Like “cyberspace,” GNSS is already here, and it has achieved broad acceptance and application. Industry and government agencies are continually enhancing technology and infrastructure and developing new GNSS-based solutions.

In this chapter, we have provided a sampling of GNSS applications and equipment, to illustrate just how incredible GNSS is. As costs are reduced further, the GNSS space segment enhanced, and new positioning techniques conceived and implemented, further applications and equipment will be developed. We are witnessing the early days of the GNSS revolution. GNSS technology will become ubiquitous – a prevalent, taken-for-granted technology in everything we do.



Afterword

“There are very few human beings who receive the truth, complete and staggering, by instant illumination. Most of them acquire it fragment by fragment on a small scale, by successive developments, cellularly, like a laborious mosaic.” Anaïs Nin, French author.

We hope reading this book has provided you with a good understanding of GNSS concepts, constellations, applications and equipment. It is a sophisticated, complex industry, so don't worry if you have not been able to absorb everything the first time through.

As Anaïs Nin so eloquently suggests, learning about GNSS may take further study, in as much depth and breadth as you need or want. For those of you interested in learning more, we have included a list of references in the Appendix, to extend your knowledge of GNSS, and to keep you informed and up to date as GNSS services, technology, applications and equipment evolve.

Like everyone else, we do not have a crystal ball. It is difficult to predict where GNSS is going to take us. The only prediction we can safely make is that the journey will likely be as incredible and exciting as it has been for the past thirty years.



Appendices

This section includes the following appendices, which include general or supplementary information about GNSS:

- Appendix A – Acronyms
- Appendix B – GNSS Glossary
- Appendix C – Standards and References



Appendix A – Acronyms

1PPS	One Pulse Per Second	C/No	Post Correlation Carrier to Noise Ratio in dB-Hz
2-D or 2D	Two Dimensional	CoCom	Coordinating Committee on Multilateral Export Controls
3-D or 3D	Three Dimensional	COG	Course Over Ground
AC	Alternating Current	COGO	Coordinate Geometry
A/D	Analog-to-Digital	COSPAS	Cosmitcheskaja Sistema Poiska Awarinitsch Sudow (Russian: space system for search of vessels in distress)
ADC	Analog-to-Digital Convertor		
ADR	Accumulated Doppler Range	CPLD	Complex Programmable Logic Device
ADR	Accumulated Delta Range	CPU	Central Processing Unit
AGC	Automatic Gain Control	CR	Carriage Return
AK	Authentication Key	CRC	Cyclic Redundancy Check
AL	Alarm Limit	CRR	Common Reference Receiver
AltBOC	Alternate Binary Offset Carrier	CS	Commercial Service
AMSAT	American Satellite	CSA	Canada Shipping Act
APC	Aircraft Power Conditioner	CSIC	Coordination Scientific Information Center
ARNS	Aeronautical Radio Navigation Services	CTP	Conventional Terrestrial Pole
ARP	Antenna Reference Point	CTS	Clear To Send
AS	Anti-Spoofing	CTS	Conventional Terrestrial System
ASCII	American Standard Code for Information Interchange	CW	Continuous Wave
ASIC	Application Specific Integrated Circuits	dB	Decibel
AVL	Automatic Vehicle Location	dBm	Decibel Relative to 1 milliwatt
BCD	Binary Coded Decimal	DC	Direct Current
BDE	Borland Database Engine	DCD	Data Carrier Detected
BDS	Black Diamond System	DCE	Data Communications Equipment (Modem)
BFS	Broadband Fiber Source	DCO	Digitally Controlled Oscillator
BIH	Bureau l'International de l'Heure	DDS	Direct Digital Sampling
BIST	Built-In Self-Test	DGNSS	Differential Global Navigation Satellite System
BIT	Built-In Test	DGPS	Differential Global Positioning System
BNR	Binary Numerical Representation	DHCP	Dynamic Host Configuration Protocol
BOC	Binary Offset Carrier	DL	Data Logger
bps	Bits per Second	DLL	Delay Lock Loop
BPSK	Bi-Phase Shift Key	DoD	Department of Defence (U.S.)
BSG	Baseband Signal Generator	DOP	Dilution Of Precision
BTS	Conventional Terrestrial System (BIH defined)	DPB	Digital Pulse Blanking
BW	Bandwidth	DR	Dead Reckoning
C/A Code	Coarse/Acquisition Code	DRAM	Dynamic Random Access Memory
CAN	Controller Area Network	DRMS	Distance Root Mean Square
CASM	Coherent Adaptive Subcarrier Modulation	DSP	Digital Signal Processor
CBIT	Continuous Built In Test	DSR	Data Set Ready
cc	Cubic Centimetres	DTE	Data Terminal Equipment
CCITT	Command, Control, and Intelligence Technical Test	DTR	Data Terminal Ready
CD	Clock Drift	D/U	Desired/Undesired
CD	Compact Disc	e	Eccentricity
cd	Change Directory	E-L	Early to Late
CDGPS	Canada-Wide Differential GPS	ECEF	Earth-Centred-Earth-Fixed
CDMA	Code Division Multiple Access	EDM	Electronic Distance Measuring (instrument)
CDPD	Cellular Digital Packet Data	EEPROM	Electrically Erasable Programmable Read Only Memory
CDU	Control and Display Unit	EGNOS	European Geostationary Navigation Overlay System
CE	Conformité Européenne (Also known as CE Mark)	EIA	Electronic Industries Alliance
CEP	Circular Error Probable	EMC	Electromagnetic Compatibility
CF	Compact Flash	EMI	Electromagnetic Immunity
CFGP	Configuration Parameters	EP	Engineering Practice
CISPR	International Special Committee On Radio Interference	ESA	European Space Agency
CKSC	Clock/Status Card	ESD	Electrostatic Discharge
CL	Long Code	ESN	Electronic Serial Number
CLK	System Clock	FAA	Federal Aviation Administration
CM	Moderate Length Code	FCC	Federal Communication Commission
CMG	Course Made Good	FDA	Frequency Distribution Amplifier
CMP	Comparator Message Processor	FDMA	Frequency Division Multiple Access
CMR	Compact Measurement Record	FEC	Forward Error Correction

FEFROM	Flash Erasable Programmable Read Only Memory	IEEE	Institute of Electrical & Electronics Engineers
FIFO	First In, First Out	IERS	International Earth Rotation Service
FKP	Flächen Korrektur Parameter (Plane Correction Parameter) German	IF	Intermediate Frequency
FLL	Frequency Lock Loop	IGP	Ionospheric Grid Point
FMEA	Failure Mode Effects Analysis	IGRF	International Geomagnetic Reference Field
FOC	Full Operational Capability	IGS CB	International GNSS Service Central Bureau
FOG	Fibre Optic Gyro	IM	Intermodulation
FOM	Figure of Merit	IMLA	Integrated Multipath Limiting Antenna
FPGA	Field-Programmable Gate Array	IMO	International Maritime Organization
FR	Factory Reset	IMU	Inertial Measuring Unit
FTP	File Transfer Protocol	INH	Inhibit
FTS	Frequency and Time Standard	INS	Inertial Navigation System
FW	Firmware	I/O	Input/Output
GAGAN	GNSS Aided GEO Augmented Navigation (India)	IODE	Issue of Data (Ephemeris)
Galn	Galileo Industries	IOV	In-Orbit Validation
GCC	Galileo Control Centre	IP	Internet Protocol
GDOP	Geometric Dilution Of Precision	IRQ	Interrupt Request
GEO	Geostationary Satellite	ISG	IF Signal Generator
GIC	GNSS Integrity Channel	ISO	International Organization for Standardization (not an acronym but a short form)
GL	GLONASS (NMEA talker ID)	ITRF	International Terrestrial Reference System
GLONASS	Global Navigation Satellite System	JPL	Jet Propulsion Laboratory (NASA)
GM	Gauss-Markov	JTAG	Joint Test Action Group
GMS	Ground Mission Segment	KPA	Klystron Power Amplifier
GMT	Greenwich Mean Time	Kb	Kilobit
GN	Combined GNSS and GLONASS (NMEA talker ID)	KB	Kilobyte
GND	Ground	KMF	Key Management Facility
GNSS	Global Navigation Satellite System	L1	The 1575.42 MHz GPS carrier frequency including C/A and P-Code
G.P.	Ground Plane	L1C	Future GPS L1 civilian frequency
GP	GNSS (NMEA talker ID)	L1F	Future Galileo L1 civilian frequency
GPAI	General Purpose Analog Input	L2	The 1227.60 MHz 2nd GPS carrier frequency (P Code only)
GPS	Global Positioning System	L2C	The L2 civilian code transmitted at the L2 frequency (1227.6 MHz)
GRAS	Ground-based Regional Augmentation System (Australia)	L5	The 1176.45 MHz 3rd civil GPS frequency that tracks carrier at low signal-to-noise ratios
GRC	Galileo Reception Chain	LAAS	Local Area Augmentation System
GRCN	Galileo Reception Chain Non-PRS	LCD	Liquid Crystal Display
GSS	Galileo Sensor Stations	LED	Light-Emitting Diode
GSTB	Galileo System Test Bed	LF	Line Feed
GTR	Galileo Test Receiver	LGF	LAAS Ground Facility
GTS	Galileo Test Signal Generator	LHCP	Left Hand Circular Polarization
GUI	Graphical User Interface	LME	Line Maintenance Equipment
GUS	Ground Uplink Station	LNA	Low Noise Amplifier
GUST	WAAS GUS-Type 1	LO	Local Oscillator
GUSTR	WAAS GUST Type-1 Receiver	LRU	Line Replacement Unit
HDOP	Horizontal Dilution Of Precision	LSB	Least significant bit
hex	Hexadecimal	LVTTTL	Low Voltage Transistor Transistor Logic
HFOM	Horizontal Figure of Merit	MAC	Media Access Control (Ethernet)
HMAC	Hashed Message Authentication Code	MAT	Multipath Assessment Tool
HP	High Performance (standard OmniSTAR service)	Mb	Megabit
HTDOP	Horizontal Position and Time Dilution Of Precision	MB	Megabyte
Hz	Hertz	mBOC	Multiplexed Binary Offset Carrier
I and Q	In-Phase and Quadrature (Channels)	MEDLL	Multipath Estimating Delay Lock Loop
I Channel	In-phase Data Channel	MEO	Medium Earth Orbit
IBIT	Initiated Built In Test	MET	Multipath Elimination Technology
IC	Integrated Circuit	Met	Meteorological
ICAO	International Civil Aviation Organization	MGRS	Military Grid Reference System
ICD	Interface Control Document		
ICP	Integrated Carrier Phase		
ID	Identification		
IEC	International Electrotechnical Commission		

Appendix A – Acronyms

MHz	MegaHertz	PPM	Parts Per Million
MIB	Management Information Base	PPP	Point to Point Protocol
MIL	Military	PPS	Precise Point Positioning Service
MINOS	Multiple Independent NOmadic Stargazer	PPS	Pulse Per Second
MKI	Mark Input	PRC	Pseudorange Correction
MKMF	Mission Key Management Facility	PRN#	PseudoRandom Noise Number
MMCX	Multimedia Communications Exchange (Lucent)	PRS	Public Regulated Service
MMT	Multipath Mitigation Technology	PSN	Part Serial Number
MOPS	Minimum Operational Performance Standard	PSR	Pseudorange
MP	Message Processor	PV	Position Valid
MPC	Modulated Precision Clock	PVT	Position Velocity Time
MPM	Multipath Meter	PZ90	Parametry Zemli 1990 (see PE-90)
MRTCA	Modified RTCA	Q Channel	Quadrature Data-Free Channel
ms	Millisecond	RAM	Random Access Memory
MSAS	MTSAT Satellite Based Augmentation System (Japan)	RAS	Remote Access Service
MSAT	Mobile Satellite	RCC	Rescue Coordination Centre
MSB	Most significant bit	RF	Radio Frequency
MSL	Mean sea level	RFDC	Radio Frequency Direct Current
MSR	Measure Output	RFU	Radio Frequency Uplink
MTBF	Mean Time Between Failures	RHCP	Right Hand Circular Polarization
MTSAT	Multi-Functional Transport Satellite	RI	Ring Indicator
N/A	Not Applicable	RINEX	Receiver Independent Exchange Format
NAS	National Airspace System (U.S.)	RLG	Ring Laser Gyro
NASA	National Aeronautics and Space Administration (U.S.)	RoHS	Restriction of the use of Hazardous Substances
NTS	National Topographic Series (Canada)	RM	Raw Measurements
NAV	RINEX Ephemeris File	ROM	Read Only Memory
NAVSTAR	NAVigation Satellite Timing And Ranging (synonymous with GPS)	RMA	Return Material Authorization
N/C or NC	Not Connected	RMS	Root Mean Square
NCC	Network Control Center (OmniSTAR)	RPDP	Relative PDP
NCO	Numerically Controlled Oscillator	RRC	Range Rate Correction
ND	Navigation Data	RSS	Residual Solution Status
NH	Neuman-Hoffman	RTC	Real-Time Clock
NMEA	National Marine Electronics Association	RTCA	Radio Technical Commission for Aviation Services
NM	Nautical mile	RTCM	Radio Technical Commission for Maritime Services
NOC	Network Operations Center	RTCMV3	RTCM Version 3.0
ns	Nanosecond	RTK	Real Time Kinematic
NVM	Non-Volatile Memory	RTS	Request To Send
OBS	RINEX Observation File	RXD	Received Data
OCXO	Oven Controlled Crystal Oscillator	SA	SMART Antenna
OEM	Original Equipment Manufacturer	SA	Selective Availability
OP	Operational Parameters	SAE	Society of Automotive Engineers
OS	Open Service	SAR	Search and Rescue
PAC	Pulsed Aperture Correlator	SARSAT	Search and Rescue Satellite Aided Tracking
PC	Personal Computer	SBAS	Satellite Based Augmentation System
PCO	Phase Centre Offset	SC	Safety Computer
P-Code	Precise Code	SCAT-I	Special Category I
PBIT	Power-Up Built-In Test	SD	Standard Deviation
PCB	Printed Circuit Board	SEP	Spherical Error Probable
PCMCIA	Personal Computer Memory Card International Association	SG	Signal Generator
PDC	Power and Data Card	SGS-90	Soviet Geodetic System 1990
PDF	Power Distribution Function	SI	Système Internationale
PDF	Portable Document File	SigGen	WAAS GUS Type-1 Signal Generator
PDP	Position Dilution Of Precision	SiS	Signal in Space
PDP	Pseudorange/Delta-Phase	SLIP	Serial Line Internet Protocol
PE-90	Parameters of the Earth 1990 (see PZ90)	SNAS	Satellite Navigation Augmentation System (China)
PIN	Position Indicator	SNR	Signal-to-Noise Ratio
PLL	Phase Lock Loop	SOL	Safety-of-Life
		SPS	Standard Position Service
		sps	Symbols Per Second

SPAN	Synchronized Position Attitude Navigation
SQM	Signal Quality Monitoring
SRAM	Static Random Access Memory
SS II	SUPERSTAR II
SU	(former) Soviet Union (now Russia)
SV	Space Vehicle
SVID	Space Vehicle Identifier
SVN	Space Vehicle Number
SW	Software
SWRU	Software Replacement Unit
TCP	Transmission Control Protocol
TCXO	Temperature Compensated Crystal Oscillator
TDOP	Time Dilution Of Precision
TES	Time Estimator Status
TIL	Time Integrity Limit
TNM	Telecommunications Network Management
TOA	Time of Almanac
TOE	Time of Ephemeris
TOW	Time of Week
TRAIM	Time Receiver Autonomous Integrity Monitor
TTFF	Time-To-First-Fix
TTL	Transistor-Transistor Logic
TTNL	Time to Narrow Lane
TVS	Transient Voltage Suppressor
TXD	Transmitted Data
UART	Universal Asynchronous Receiver Transmitter
UDP	User Datagram Protocol
UDRE	User Differential Range Error
UHF	Ultra High Frequency
USB	Universal Serial Bus
USGS	United States Geological Survey
UTC	Coordinated Universal Time
UTC(SU)	Coordinated Universal Time (former Soviet Union, now Russia)
V AC	Volts Alternating Current
V DC	Volts Direct Current
VARF	Variable Frequency
VBS	Virtual Base Station (standard OmniSTAR service)
VCTCXO	Voltage Controlled Temperature Compensated Crystal Oscillator
VDOP	Vertical Dilution of Precision
VFD	Vacuum Fluorescent Display
VFOM	Vertical Figure of Merit
VSWR	Voltage Standing Wave Ratio
WAAS	Wide Area Augmentation System
WAAS G-II	WAAS Reference Receiver: G-II
WADGNSS	Wide Area DGNSS
WEEE	Waste Electrical and Electronic Equipment
WGS	World Geodetic System
WHQL	Windows Hardware Quality Lab (Microsoft)
WMP	WAAS Message Processor
WNA	Week number of almanac
WPT	Waypoint
XP	Extra Performance (standard OmniSTAR service)
XTE	Crosstrack Error
ZUPT	Zero Velocity Update

Absolute Accuracy

In GNSS positioning, absolute accuracy is the degree to which the position of an object on a map conforms to its correct location on the earth according to an accepted coordinate system.

Acquisition

The process of locking onto a satellite's C/A code and P-code. A receiver acquires all available satellites when it is first powered up, then acquires additional satellites as they become available and continues tracking them until they become unavailable.

Address Field

For sentences in the NMEA standard, the fixed length field following the beginning sentence delimiter "\$" (HEX 24). For NMEA approved sentences, composed of a two character talker identifier and a three character sentence formatter. For proprietary sentences, composed of the character "P" (HEX 50) followed by a three character manufacturer identification code.

ADR

Accumulated Doppler Range. Carrier phase, in cycles.

Almanac

A set of orbit parameters that allows calculation of approximate GNSS satellite positions and velocities. The almanac is used by a GNSS receiver to determine satellite visibility and as an aid during acquisition of GNSS satellite signals.

Almanac Data

A set of data which is downloaded from each satellite over the course of 12.5 minutes. It contains orbital parameter approximations for all satellites, GNSS to universal standard time (UTC) conversion parameters, and single-frequency ionospheric model parameters.

Ambiguity

The integer number of carrier cycles between a satellite and receiver.

Anti-Spoofing

Denial of the P-code by the Control Segment is called Anti-Spoofing. It is normally replaced by encrypted Y-code. [See "*P-Code*" and "*Y-Code*"]

Antipodal Satellites

Antipodal satellites are satellites in the same orbit plane separated by 180 degrees in argument of latitude.

ASCII

A 7-bit wide serial code describing numbers, upper and lower case characters, special and non-printing characters. Typically used for textual data.

Attenuation

Reduction of signal strength

Azimuth

The horizontal direction of a celestial point from a terrestrial point, expressed as the angular distance from 000° (reference) clockwise through 360°. The reference point is generally True North, but may be Magnetic North, or Relative (ship's head).

Baseline

The line between a pair of stations for which simultaneous GNSS data has been collected.

Base Station

The GNSS receiver which is acting as the stationary reference. It has a known position and transmits messages for the rover receiver to use to calculate its position.

Bearing

The horizontal direction of one terrestrial point from another terrestrial point, expressed as the angular distance from a reference direction, usually measured from 000° at the reference direction clockwise through 360°. The reference point may be True North, Magnetic North, or Relative (ship's head).

Bluetooth

A short-range radio technology for Internet and mobile devices, aimed at simplifying communications among them.

Broadcast Ephemerides

A set of parameters which describes the location of satellites with respect to time, and which is transmitted (broadcast) from the satellites.

Canada-Wide Differential Global Positioning System (CDGPS)

The CDGPS system is a free Canada-wide DGPS service that is accessible coast-to-coast, throughout most of the continental United States, and into the Arctic.

Carrier

The steady transmitted RF signal whose amplitude, frequency, or phase may be modulated to carry information.

Carrier Phase Ambiguity

The number of integer carrier phase cycles between the user and the satellite at the start of tracking. (Sometimes ambiguity for short).

Carrier Phase Measurements

These are “Accumulated Doppler Range” (ADR) measurements. They contain the instantaneous phase of the signal (modulo 1 cycle) plus some arbitrary number of integer cycles. Once the receiver is tracking the satellite, the integer number of cycles correctly accumulates the change in range seen by the receiver. When a “lock break” occurs, this accumulated value can jump an arbitrary integer number of cycles (this is called a cycle slip).

C-Band

C Band is the original frequency allocation for communications satellites. C-Band uses 3.7-4.2 GHz for downlink and 5.925-6.425 GHz for uplink.

Checksum

By NMEA standard, a validity check performed on the data contained in the sentences, calculated by the talker, appended to the message, then recalculated by the listener for comparison to determine if the message was received correctly. Required for some sentences, optional for all others.

Circular Error Probable (CEP)

The radius of a circle such that 50% of a set of events occur inside the boundary.

Coarse Acquisition (C/A) Code

A pseudorandom string of bits that is used primarily by commercial GNSS receivers to determine the range to the transmitting GNSS satellite. The 1023 chip GPS C/A code repeats every 1 ms giving a code chip length of 300 m, which is very easy to lock onto.

Communication Protocol

A method established for message transfer between a talker and a listener which includes the message format and the sequence in which the messages are to be transferred. Also includes the signalling requirements such as bit rate, stop bits, parity, and bits per character.

Control Point

[See “*Ground Control Point (GRP)*”]

Control Segment

The Master Control Station and the globally dispersed Reference Stations used to manage the GNSS satellites, determine their precise orbital parameters, and synchronize their clocks.

Controller Area Network Bus (CAN Bus)

A rugged serial bus with a protocol that provides services for processes, data and network management.

Coordinated Universal Time

This time system uses the second-defined true angular rotation of the Earth measured as if the Earth rotated about its Conventional Terrestrial Pole. However, UTC is adjusted only in increments of one second. The time zone of UTC is that of Greenwich Mean Time (GMT).

Course

The horizontal direction in which a vessel is to be steered or is being steered; the direction of travel through the air or water. Expressed as angular distance from reference North (either true, magnetic, compass, or grid), usually 000° (north), clockwise through 360°. Strictly, the term applies to direction through the air or water, not the direction intended to be made good over the ground [See "*Track Made Good*"]. Differs from heading.

Course Made Good (CMG)

The single resultant direction from a given point of departure to a subsequent position; the direction of the net movement from one point to the other. This often varies from the track caused by inaccuracies in steering, currents, cross-winds, etc. This term is often considered to be synonymous with Track Made Good, however, Course Made Good is the more correct term.

Course Over Ground (COG)

The actual path of a vessel with respect to the Earth (a misnomer in that courses are directions steered or intended to be steered through the water with respect to a reference meridian); this will not be a straight line if the vessel's heading yaws back and forth across the course.

Cross Track Error (XTE)

The distance from the vessel's present position to the closest point on a great (XTE) Circle line connecting the current waypoint coordinates. If a track offset has been specified in the receiver SETNAV command, the cross track error will be relative to the offset track great circle line.

Cycle Slip

When the carrier phase measurement jumps by an arbitrary number of integer cycles. It is generally caused by a break in the signal tracking due to shading or some similar occurrence.

Dead Reckoning

The process of determining a vessel's approximate position by applying (DR) from its last known position a vector or a series of consecutive vectors representing the run that has since been made, using only the courses being steered, and the distance run as determined by log, engine rpm, or calculations from speed measurements.

Destination

The immediate geographic point of interest to which a vessel is navigating. It may be the next waypoint along a route of waypoints or the final destination of a voyage.

Differential GNSS (DGNSS)

A technique to improve GNSS accuracy that uses pseudorange errors at a known location to improve the measurements made by other GNSS receivers within the same general geographic area.

Dilution of Precision (DOP)

A numerical value expressing the confidence factor of the position solution based on current satellite geometry. The lower

the value, the greater the confidence in the solution. DOP can be expressed in the following forms.

- GDOP: Uncertainty of all parameters (latitude, longitude, height, clock offset)
- PDOP: Uncertainty of 3-D parameters (latitude, longitude, height)
- HDOP: Uncertainty of 2-D parameters (latitude, longitude, time)
- VDOP: Uncertainty of height parameter
- TDOP: Uncertainty of clock offset parameter

Doppler

The change in frequency of sound, light, or other wave caused by movement of its source relative to the observer.

Theoretical Doppler: The expected Doppler frequency based on a satellite's motion relative to the receiver. It is computed using the satellite's coordinates and velocity, and the receiver's coordinates and velocity.

Apparent Doppler: Same as Theoretical Doppler of satellite above, with clock drift correction added.

Instantaneous Carrier: The Doppler frequency measured at the receiver, at that epoch.

Doppler Aiding

A signal processing strategy, which uses a measured Doppler shift to help a receiver smoothly track the GNSS signal, to allow more precise velocity and position measurement.

Double-Difference

A mathematical technique comparing observations by differencing between receiver channels and then between the base and rover receivers.

Double-Difference Carrier Phase Ambiguity

Carrier phase ambiguities which are differenced between receiver channels and between the base and rover receivers.

They are estimated when a double-difference mechanism is used for carrier phase positioning. (Sometimes double-difference ambiguity or ambiguity, for short).

Earth-Centred-Earth-Fixed (ECEF)

This is a coordinate system which has the X-axis in the Earth's equatorial plane pointing to the Greenwich prime meridian, the Z-axis pointing to the north pole, and the Y-axis in the equatorial plane 90° from the X-axis with an orientation which forms a right-handed XYZ system.

Eccentricity (e)

A dimensionless measurement defined for a conic section where $e=0$ is a circle, $0 < e < 1$ is an ellipse, $e=1$ is a parabola and $e > 1$ is a hyperbola. For an ellipse, larger values of e correspond to a more elongated shape. The eccentricity of GNSS satellite orbit is typically .02.

Elevation

The angle from the horizon to the observed position of a satellite.

Ellipsoid

A smooth mathematical surface which represents the Earth's shape and very closely approximates the geoid. It is used as a reference surface for geodetic surveys.

Ellipsoidal Height

Height above a defined ellipsoid approximating the surface of the Earth.

Ephemeris

A set of satellite orbit parameters that are used by a GNSS receiver to calculate precise GNSS satellite positions and velocities. The ephemeris is used in the determination of the navigation solution and is updated periodically by the satellite to maintain the accuracy of GNSS receivers.

Ephemeris Data

The data downlinked by a GNSS satellite describing its own orbital position with respect to time.

Epoch

Strictly a specific point in time. Typically when an observation is made.

Field

A character or string of characters immediately preceded by a field delimiter.

Fixed Ambiguity Estimates

Carrier phase ambiguity estimates which are set to a given number and held constant. Usually they are set to integers or values derived from linear combinations of integers.

Fixed Discrete Ambiguity Estimates

Carrier phase ambiguities which are set to values that are members of a predetermined set of discrete possibilities, and then held constant.

Fixed Field

A field in which the number of characters is fixed, including the cyclic redundancy check (CRC) field.

Fixed Integer Ambiguity Estimates

Carrier phase ambiguities which are set to integer values and then held constant.

Flash ROM

Programmable read-only memory.

Galileo

Galileo will be the European Union's own global navigation satellite system, providing a highly accurate, guaranteed global positioning service under civilian control. The fully deployed Galileo system will consist of 30 satellites (27 operational + 3 active spares), positioned in three circular orbits, 23616 km above the Earth, and at an inclination of the orbital planes of 56 degrees with reference to the equatorial plane.

Galileo Industries (Galn)

Galn is a consortium of European prime companies charged with the development of the Galileo system for the European Space Agency.

Geometric Dilution of Precision (GDOP)

[See "*Dilution of Precision (DOP)*"]

Geoid

The shape of the Earth if it were considered as a sea level surface extended continuously through the continents. The geoid is an equipotential surface coincident with mean sea level to which at every point the plumb line (direction in which gravity acts) is perpendicular. The geoid, affected by local gravity disturbances, has an irregular shape.

Geodetic Datum

The reference ellipsoid surface that defines the coordinate system.

Geo-stationary

A satellite orbit along the equator that results in a constant fixed position over a particular reference point on the Earth's surface.

Global Navigation Satellite System (GLONASS)

GLONASS is a radio satellite navigation system, the Russian counterpart to the United States' GPS and the European Union's Galileo positioning systems. When complete, the GLONASS space segment will consist of 24 satellites in three orbital planes,

with eight satellites per plane in three orbital planes. The satellites are placed into nominally circular orbits with target inclinations of 64.8 degrees and an orbital height of about 19,140 km, which is about 1,050 km lower than GPS satellites.

Global Positioning System (GPS)

Full name is NAVSTAR Global Positioning System. A space-based radio positioning system which provides suitably equipped users with accurate position, velocity and time data. GPS provides this data free of direct user charge worldwide, continuously, and under all weather conditions. The GPS constellation consists of a minimum of 24 orbiting satellites, four equally spaced around each of six different orbital planes. The system is being developed by the Department of Defense under U.S. Air Force management.

GPS L1 Frequency

The 1575.42 MHz GPS carrier frequency, which contains the course acquisition (C/A) code, as well as encrypted P-code, and navigation messages used by commercial GPS receivers.

GPS L2 Frequency

The 1227.60 MHz secondary GPS carrier frequency, containing only encrypted P-code. Currently, GPS satellites transmit the civilian C/A code on the L1 frequency, and the military P(Y) code on both the L1 and L2 frequencies. New Block IIR-M GPS satellites will transmit the same signals as previous GPS satellites, but will also have a new signal, called L2C, on the L2 frequency.

GPS L5 Frequency

The third civil GPS frequency at 1176.45 MHz beginning with the first Block IIF NAVSTAR GPS satellite to be launched in 2009. This frequency is located within the 960-1215 MHz frequency band. The L5 signal is equally split between an in-phase (I) data channel and a quadrature (Q) data-free channel, which improves resistance to interference, especially from pulse emitting systems in the same band as L5.

Great Circle

The shortest distance between any two points along the surface of a sphere or ellipsoid, and therefore the shortest navigation distance between any two points on the Earth. Also called Geodesic Line.

Handshaking

Predetermined hardware or software activity designed to establish or maintain two machines or programs in synchronization. Handshaking concerns the exchange of messages or packets of data between two systems with limited buffers. Hardware handshaking uses voltage levels or pulses in wires to carry the handshaking signals. Software handshaking uses data units (for example, binary bits) carried by some underlying communication medium.

Heading

The direction in which a vessel points or heads at any instant, expressed in degrees 000° clockwise through 360° and may be referenced to True North, Magnetic North, or Grid North. The heading of a vessel is also called the ship's head. Heading is a constantly changing value as the vessel oscillates or yaws across the course due to the effects of the air or sea, cross currents, and steering errors.

Horizontal Dilution of Precision (HDOP)

[See "*Dilution of Precision (DOP)*"]

Horizontal and Time Dilution of Precision (HTDOP)

[See "*Dilution of Precision (DOP)*"]

Integer Ambiguity Estimates

Carrier phase ambiguity estimates which are only allowed to take on integer values.

Iono-Free Carrier Phase Observation

A linear combination of L1 and L2 carrier phase measurements which provides an estimate of the carrier phase observation on one frequency with the effects of the ionosphere removed. It provides a different ambiguity value (non-integer) than a simple measurement on that frequency.

Kinematic

The user's GNSS antenna is moving. In GNSS, this term is typically used with precise carrier phase positioning, and the term dynamic is used with pseudorange positioning.

L-band

L-band is a frequency range between 390 MHz and 1.55 GHz which is used for satellite communications and for terrestrial communications between satellite equipment. L-band includes the GNSS carrier frequencies L1, L2, CDGPS and the Omni-STAR satellite broadcast signal.

Lane

A particular discrete ambiguity value on one carrier phase range measurement or double-difference carrier phase observation. The type of measurement is not specified (L1, L2, L1-L2, iono-free).

Local Observation Set

An observation set taken by the receiver on which the software is operating.

Local Tangent Plane

A coordinate system based on a plane tangent to the ellipsoid's surface at the user's location. The three coordinates are east, north and up. Latitude, longitude and height positions operate in this coordinate system.

Low-Latency Solution

A position solution which is based on a prediction. A model (based on previous base station observations) is used to estimate what the observations will be at a given time epoch. These estimated base station observations are combined with actual measurements taken at the rover station to provide a position solution.

Magnetic Bearing

Bearing relative to magnetic north; compass bearing corrected for deviation.

Magnetic Heading

Heading relative to magnetic north.

Magnetic Variation

The angle between the magnetic and geographic meridians at any place, expressed in degrees and minutes east or west to indicate the direction of magnetic north from true north.

Mask Angle

The minimum GNSS satellite elevation angle permitted by a particular receiver design. Satellites below this angle will not be used in position solution.

Matched Observation Set Pair

Observations from both the base station and the local receiver which have been matched by time epoch, contain the same satellites, and are corrected for any known offsets.

Measurement Error Variance

The square of the standard deviation of a measurement quantity. The standard deviation is representative of the error typically expected in a measured value of that quantity.

Measurement Time Epoch

The point in time at which a receiver takes a measurement.

Misclosure

The gap between a receiver's predicted and actual position.

Moving Base Station

The GNSS receiver which is acting as the reference point but is in motion. It has an estimated position and transmits messages for the rover receiver to use to calculate its position.

Multipath Errors

GNSS positioning errors caused by the interaction of the satellite signal and its reflections.

Nanosecond

1×10^{-9} second.

Narrow Lane

The GPS observable obtained by summing the carrier-phase observations on the L1 and L2 frequencies. The narrow-lane observable can help resolve carrier-phase ambiguities.

Network RTK

With Network RTK, corrections are generated from a base station network instead of from a single base station. These corrections can remove more spatially correlated errors and thus improve the RTK performance as opposed to the traditional RTK. Network RTK uses permanent base station installations, allowing kinematic GNSS users to achieve centimetre accuracies without the need for setting up a GNSS base station on a known site.

Non-Volatile Memory

A type of memory device that retains data in the absence of a power supply.

Null Field

By NMEA standard, indicates that data is not available for the field. Indicated by two ASCII commas, for example, “,” (HEX 2C2C), or, for the last data field in a sentence, one comma followed by either the checksum delimiter “*” (HEX 2A) or the sentence delimiters <CR><LF> (HEX 0D0A). [Note: the ASCII Null character (HEX 00) is not to be used for null fields.]

Obscuration

Term used to describe periods of time when a GNSS receiver's line-of-sight to GNSS satellites is blocked by natural or man-made objects.

Observation

Any measurement.

Observation Set

A set of receiver measurements taken at a given time which includes one time for all measurements, and the following for each satellite tracked: PRN number, pseudorange or carrier phase or both, lock time count, signal strength, and tracking status.

OmniSTAR

A wide-area GNSS correction service, using L-band satellite broadcast frequencies (1525 - 1560 MHz). Data from many widely-spaced Reference Stations is used in a proprietary multi-site solution. OmniSTAR Virtual Base Station (VBS) types achieve sub-metre positioning over most land areas worldwide while OmniSTAR High Performance (HP) types achieve 10 cm accuracy. Use of the OmniSTAR service requires a subscription.

Origin Waypoint

The starting point of the present navigation leg, expressed in latitude and longitude.

Parallel Receiver

A receiver that monitors four or more satellites simultaneously with independent channels.

Parity

The even or odd quality of the number of ones or zeroes in a binary code. Parity is often used to determine the integrity of data especially after transmission.

Perigee

The point in a body's orbit at which it is nearest the Earth.

P-Code

Precise code or protected code. A pseudorandom string of bits that is used by GPS receivers to determine the range to the transmitting GPS satellite. P-code is replaced by an encrypted Y-code when Anti-Spoofing is active. Y-code is intended to be available only to authorized (primarily military) users. [See *“Anti-Spoofing”*, *“(C/A) Code”* and *“Y-Code”*]

PDOP

Position Dilution of Precision [See *“Dilution of Precision (DOP)”*]

Post-processing

A processing mode in which a base station is placed at a known reference point and a rover is used for gathering positions. Accurate coordinates are generated by taking data stored from the receivers and processing them using post-processing software.

Precise Positioning Service (PPS)

The GNSS positioning, velocity, and time service which is available on a continuous, worldwide basis to users authorized by the U.S. Department of Defense (typically using P-code).

PRN Number

A number assigned by the GPS system designers to a given set of pseudorandom codes. Typically, a particular satellite will keep its PRN (and hence its code assignment) indefinitely, or at least for a long period of time. It is commonly used as a way to label a particular satellite.

Pseudolite

An Earth-based transmitter designed to mimic a satellite.

Pseudorange

The calculated range from the GNSS receiver to the satellite determined by taking the difference between the measured satellite transmit time and the receiver time of measurement, and multiplying by the speed of light. Contains several sources of error.

Pseudorange Measurements

Measurements made using one of the pseudorandom codes on the GNSS signals. They provide an unambiguous measure of the range to the satellite including the effect of the satellite and user clock biases.

PZ-90

Parametri Zemli 1990 (PZ-90, or in English translation, Parameters of the Earth 1990, PE-90) geodetic datum. GLONASS information is referenced to the PZ-90.02 geodetic datum, and GLONASS coordinates are reconciled in GLONASS-capable NovAtel receivers through a position filter and output to WGS84.

Radio Technical Commission for Aeronautics (RTCA)

An organization which developed and defined a message format for differential positioning.

Radio Technical Commission for Maritime Services (RTCM)

An organization which developed and defined the SC-104 message format for differential positioning.

Real-Time Kinematic (RTK)

A type of differential positioning based on observations of carrier phase.

Receiver Channels

A GNSS receiver specification which indicates the number of independent hardware signal processing channels included in the receiver design.

Reference Satellite

In a double-difference implementation, measurements are differenced between different satellites on one receiver in order to cancel the correlated errors. Usually one satellite is chosen as the “reference”, and all others are differenced with it.

Reference Station

[See “*Base Station*”]

Relative Bearing

Bearing relative to heading or to the vessel.

Remote Station

[See “*Rover Station*”]

Residual

In the context of measurement, the residual is the difference between the measurement predicted by the computed solution and the actual measurement.

Root Mean Square (RMS)

A probability level of 68%.

Route

A planned course of travel, usually composed of more than one navigation leg.

Rover Station

The GNSS receiver which does not know its position and needs to receive measurements from a base station to calculate differential GNSS positions. (The terms remote and rover are interchangeable.)

SafeTrak

The receiver tracks a satellite by replicating the satellite’s PRN code and aligning it with the received PRN code. A cross-correlation check is performed to check alignment and the cross-correlation channel shifts its code phase repeatedly to measure the power. If necessary, the tracking channel re-acquires the satellite to remove the cross-correlation error.

Safety-of-Life (SOL)

The safety-of-life service will be offered to Galileo users who are highly dependant on precision, signal quality and signal transmission reliability. It will offer a high level of integrity, and consequently, provide the user with a very rapid warning of any possible malfunctions.

The safety-of-life service will be transmitted in two frequency bands. On the L1 at 1575.42 MHz and on E5a+E5b at 1207.14 MHz. Users may receive signals from the two frequency bands independently.

Satellite-Based Augmentation System (SBAS)

A type of geo-stationary satellite system that improves the accuracy, integrity, and availability of the basic GNSS signals. This includes WAAS, EGNOS, and MSAS.

Selective Availability (SA)

The method used in the past by the United States Department of Defense to control access to the full accuracy achievable by civilian GPS equipment (generally by introducing timing and ephemeris errors).

Selected Waypoint

The waypoint currently selected to be the point toward which the vessel is travelling. Also called “to” waypoint, destination or destination waypoint.

Sequential Receiver

A GNSS receiver in which the number of satellite signals to be tracked exceeds the number of available hardware channels. Sequential receivers periodically reassign hardware channels to particular satellite signals in a predetermined sequence.

Sidereal Day

A sidereal day is the rotation period of the Earth relative to the equinox and is equal to one calendar day (the mean solar day) minus approximately four minutes.

Signal Quality Monitoring (SQM)

Signal Quality Monitoring (SQM) technology is used to monitor GNSS and GEO signals in space for anomalous behavior.

Spherical Error Probable (SEP)

The radius of a sphere, centred at the user’s true location, that contains 50 percent of the individual three-dimensional position measurements made using a particular navigation system.

Spheroid

Sometimes known as ellipsoid; a perfect mathematical figure which very closely approximates the geoid. Used as a surface of reference for geodetic surveys.

Standard Positioning Service (SPS)

A positioning service made available by the United States Department of Defense which is available to all GPS civilian users on a continuous, worldwide basis (typically using C/A Code).

Space Vehicle ID (SV)

Sometimes used as SVID. A unique number assigned to each satellite for identification purposes. The ‘space vehicle’ is a GNSS satellite.

TDOP

Time Dilution of Precision [See “*Dilution of Precision (DOP)*”]

Three-Dimensional Coverage (hours)

The number of hours per day when four or more satellites are available with acceptable positioning geometry. Four visible satellites are required to determine location and altitude.

Three-Dimensional (3D) Navigation

Navigation mode in which altitude and horizontal position are determined from satellite range measurements.

Time-To-First-Fix (TTFF)

The actual time required by a GNSS receiver to achieve a position solution. This specification will vary with the operating state of the receiver, the length of time since the last position fix, the location of the last fix, and the specific receiver design.

Track Made Good

The single resultant direction from a point of departure to a point of arrival or subsequent position at any given time; may be considered synonymous with Course Made Good.

True Bearing

Bearing relative to true north; compass bearing corrected for compass error.

True Heading

Heading relative to true north.

Two-Dimensional (2D) Coverage

The number of hours-per-day with three or more satellites visible. Three visible satellites can be used to determine location if the GNSS receiver is designed to accept an external altitude input.

Two-Dimensional Navigation

Navigation mode in which a fixed value of altitude is used for one or more position calculations while horizontal (2D) position can vary freely based on satellite range measurements.

Undulation

The distance of the geoid above (positive) or below (negative) the mathematical reference ellipsoid (spheroid). Also known as geoidal separation, geoidal undulation, geoidal height.

Update Rate

The GNSS receiver specification which indicates the solution rate provided by the receiver when operating normally.

UTC

[See “*Coordinated Universal Time*”]

VDOP

Vertical Dilution of Precision [See “*Dilution of Precision (DOP)*”]

Variable Field

By NMEA standards, a data field which may or may not contain a decimal point and which may vary in precision following the decimal point depending on the requirements and the accuracy of the measuring device.

Waypoint

A reference point on a track.

Wideband Antenna

A GNSS antenna that is capable of receiving multiple Global Navigation Satellite Systems (GNSS) including GPS, Galileo and GLONASS frequencies.

Wide Lane

A particular integer ambiguity value on one carrier phase range measurement or double-difference carrier phase observation when the difference of the L1 and L2 measurements is used. It is a carrier phase observable formed by subtracting L2 from L1 carrier phase data: $F' = F1 - F2$. The corresponding wavelength is 86.2 cm.

World Geodetic System 1984 (WGS84)

An ellipsoid designed to fit the shape of the entire Earth as well as possible with a single ellipsoid. It is often used as a reference on a worldwide basis, while other ellipsoids are used locally to provide a better fit to the Earth in a local region. GNSS uses the centre of the WGS84 ellipsoid as the centre of the GNSS ECEF reference frame.

Y-Code

An encrypted form of P-code. Satellites transmit Y-Code in place of P-code when Anti-Spoofing is in effect. [See “*P-Code*” and “*Anti-Spoofing*”]

This appendix provides links to companies and agencies engaged in activities related to GNSS. Web site addresses are subject to change; however, they are accurate at the time of this book's publication.

NovAtel Inc.

Contact your local NovAtel dealer first for more information. To locate a dealer in your area or to resolve a technical problem, contact NovAtel Inc. directly.

Customer Service Department

1120 - 68 Avenue NE

Calgary, AB., Canada, T2E 8S5

Phone: 1-800-NOVATEL (International), +1-403-295-4900 (U.S. & Canada)

Fax: +1-403-295-4901

E-mail: support@novatel.com

Web site: <http://www.novatel.com>

Other sources of information about GNSS:

Arinc: <http://www.arinc.com>

Canadian-Wide Differential GPS (CDGPS): <http://www.cdgps.com>

Geodetic Survey of Canada: <http://www.geod.nrcan.gc.ca/>

National Geodetic Survey: <http://www.ngs.noaa.gov>

National Marine Electronics Association (NMEA): <http://www.nmea.org>

NAVSTAR GPS Operations: <http://tycho.usno.navy.mil/gpsinfo.html>

OmniSTAR: <http://www.omnistar.com>

Radio Technical Commission for Aeronautics (RTCA): <http://www.rtca.org>

Radio Technical Commission for Maritime Services (RTCM): <http://www.rtcn.org>

Russian Space Agency (for information about GLONASS): <http://www.GLONASS-ianc.rsa.ru>

Society of Automotive Engineers (SAE): <http://www.sae.org/servlets/index>

A

- aerial photogrammetry, 56
- agriculture, 54
- almanac, 19
- ambiguity resolution, 45
- antennas, 11
- application
 - safety-of-life, 32
- applications, 11, 25, 52
- augmentation, 12
- aviation, 53

B

- base station data
 - post-processing, 49

C

- Canada-wide Differential GPS (CDGPS), 43
- carrier-based techniques, 26, 44
- CDMA, 16
- clocks, 16
- combined GPS/GLONASS mode, 35
- Compass (China), 7, 37
- computation, 22
- constellations, 15
- consumer applications, 52
- control segment, 9
- construction, 55
- corrections, 41-42

D

- data
 - post-processing, 48
- defense
 - navigation, 59
 - reconnaissance and map creation, 59
 - search and rescue, 59
 - unmanned vehicles, 59
- differential corrections 41-42
- differential GNSS, 26, 41-42
- differential positioning, 41-44
- Dilution of Precision. See DOP
- DOP, 45

E

- ephemerides, 15
- ephemeris, 18
- equipment
 - accuracy, 60
 - acquisition times, 61
 - computational requirements, 61

- data storage, 61
- description, 59
- emissions, 61
- environmental, 61
- portability, 61
- power consumption, 61
- regulatory compliance, 61
- reliability, 61
- shock and vibration, 61
- size, 61
- user interface, 61

European Geostationary Navigation Overlay Service (EGNOS), 43

F

- Free Open Service (OS), 37
- frequencies, 17

G

- GAGAN (GPS-Aided GEO Augmented Navigation System), 44
- Galileo (European Union),
 - description, 35
 - overview, 7
 - Rescue Coordination Centre (RCC), 36
 - satellites, 15
 - search and rescue (SAR), 36
 - services, 37
 - system design, 36
 - System Test Bed (GSTB), 35
- Galileo Control Centres (GCC), 36
- Galileo Sensor Stations (GSS), 36
- Geospatial Information Systems (GIS), 57
- geosynchronous orbit, 41
- GLONASS (Global Navigation Satellite System, Russia)
 - control segment, 34
 - history, 32
 - overview, 7
 - signals, 34
 - space segment, 33
 - system design, 33
- GNSS
 - architecture, 7
 - concepts, 14
 - error sources, 25
 - overview, 6
 - systems, 7
- Google Earth, 56
- Government encrypted Public Regulated Service (PRS), 37
- GPS (Global Positioning System, United States)
 - control segment, 31

- history, 29
- overview, 7
- signals, 30
- space segment, 29

GPS modernization, 31

ground mapping, 56

I

Inertial Navigation System (INS), 26, 47

ionospheric delays, 18, 20

IRNSS (Indian Regional Navigation Satellite System, India), 38

L

L1, 17, 18, 30, 34

L2, 19, 30

L2C, 31

L5, 32

LIDAR (Light Detection and Ranging), 56

L-band, 18

M

machine control

- accuracy, 54
- data management, 54
- description, 53
- efficiency, 54
- job management, 54
- theft detection, 54

marine applications, 58

marine transportation, 53

multipath, 21

N

NAVSTAR, 29-32

O

OmniSTAR, 44

operational satellites, 7

P

p(Y) code, 18, 30

port automation, 57

position,

- computation, 22

positioning

- concepts, 6, 14

propagation, 20

pseudorandom code, 17

Pseudo Random code Number (PRN), 30

pseudorange, 20

Q

QZSS (Quasi-Zenith Satellite System, Japan), 38

R

rail transportation, 53

receivers, 12

reception, 21

remote sensing, 56

RoHS, 61

S

Safety of Life, 32, 37

satellites

- clocks, 16
- corrections, 20
- description, 15
- errors, 20
- lifetimes, 20
- orbits, 16
- range, 16, 20
- signals, 16

search and rescue, 36

signals, 9

sonobuoys, 58

space segment, 8

Space Vehicle Number (SVN), 30

StarFire, 44

surface mining, 55

surface transportation, 53

surveying, 55

T

time applications, 58

time to first fix, 61

transportation, 53

trilateration, 6, 22

tropospheric delays, 21

U

user equipment, 11

user segment, 9, 21, 25

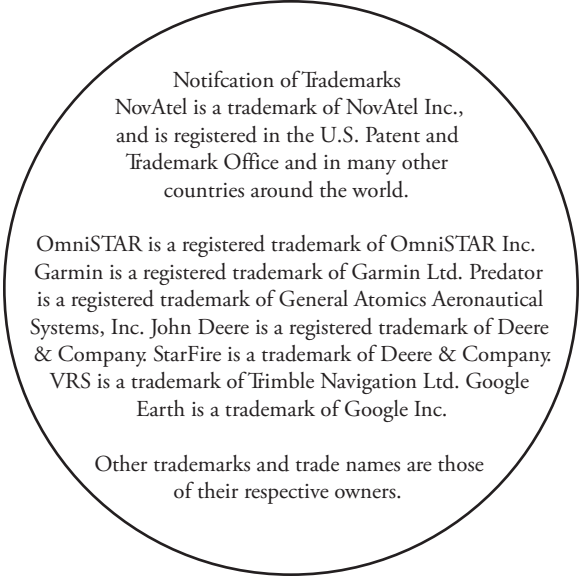
V

VERIPOS, 44

virtual reference stations (VRS), 45

W

WEEE, 61



Notification of Trademarks
NovAtel is a trademark of NovAtel Inc.,
and is registered in the U.S. Patent and
Trademark Office and in many other
countries around the world.

OmniSTAR is a registered trademark of OmniSTAR Inc.
Garmin is a registered trademark of Garmin Ltd. Predator
is a registered trademark of General Atomics Aeronautical
Systems, Inc. John Deere is a registered trademark of Deere
& Company. StarFire is a trademark of Deere & Company.
VRS is a trademark of Trimble Navigation Ltd. Google
Earth is a trademark of Google Inc.

Other trademarks and trade names are those
of their respective owners.

Printed in Canada
© Copyright 2010

ISBN 978-0-9813754-0-3

